



Bricks

A LEGO® BRICK MODEL OF

VHS Cassette

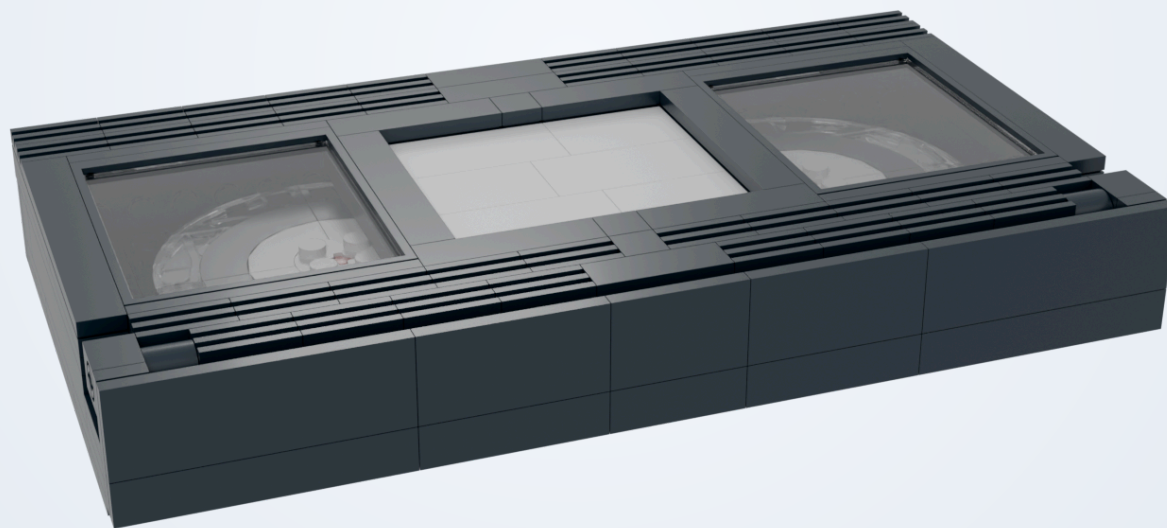
A 1:1 VHS tape, within a few millimetres of the real 187 x 103 x 25 mm cassette. Swing the dust flap up to see the tape; watch the reels turn through the windows.

384 pieces · 104 steps · 8 sections

Unofficial fan-made building instructions

Not physically build-tested. Every part is a real LEGO element and the model was checked on a computer (connections, collisions, every step, balance, mechanism), but nobody has built it with real bricks yet.

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Before you start

An unofficial fan-made manual

This booklet is not a LEGO product. It was generated from the model file by a computer program and checked part by part.

Not physically build-tested

The model and these steps were checked on a computer only: every part/colour is a real LEGO element, all 996 connections between parts were matched in 3D, no two parts overlap, every part can be pushed into place in its step, and the model's balance was estimated; the mechanism was swept through its whole range. Not tested: how firmly parts hold, part tolerances and how it copes with handling. Take your time and support the model while you work.

The finished model

About 18.4 × 10.2 cm and 2.6 cm tall, about 313 g (estimated).

Built like a real cassette, from the inside out. First the chassis: the bottom with its drive holes and screw heads, the lamp tube, the reel-lock release, the guide rollers and the tape stretched across the mouth. Two reels drop onto their red spindles: the supply reel and the take-up reel. The end walls go on, then the top shell, built upside down with its grip bands, recessed label, spine label and two big windows, is lowered over everything. Last, the full-length dust flap hangs on two Technic pins.

You will need

- The 384 pieces in 5 colours listed at the back, with their LEGO element IDs and BrickLink numbers.
- A brick separator for pieces that sit too deep for a finger.

Tips

- Sort the pieces by colour before you start.
- Build the sections in order: each sub-assembly is finished before it is used.
- New pieces in a step have a yellow outline; earlier pieces are drawn paler.
- Each reel turns on its red Technic axle in the chassis' round holes. The axle's end shows in the drive hole underneath: turn it with a fingertip to wind the tape.
- Lower the top shell straight down so the window hangers find their ledges and the reel hubs slide under the windows.
- The write-protect tab on the back is a single tile: pull it off to protect your recording.

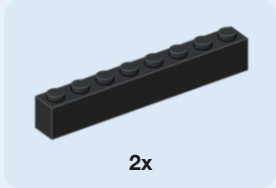
The 5 colours

- | | |
|---|--|
|  Black: 335 |  Flat Silver: 5 |
|  White: 32 |  Red: 2 |
|  Trans-Clear: 10 | |

How to read the steps

Parts and step number

7

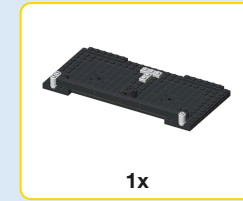


The parts for the step and how many of each, then the step number. The parts of a step are drawn at one scale, so a longer one looks longer; a tile or a long plate also has its size in studs under it.

New parts

The parts added in a step have a yellow outline and are drawn in full colour; the parts from earlier steps are drawn paler.

Sub-assemblies



A yellow-framed picture means: put in the sub-assembly you built in its own section. "Build 4x" means you need four of them.

Sections

Each section starts with its name at the top of the page. Sections are built in the order they appear.

Seen from below

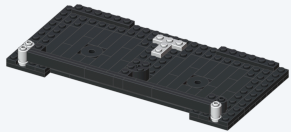
When a step adds parts underneath, the picture shows the model from below. A small label says so (and "seen from behind" likewise). A piece the picture cannot show gets a small picture beside it, ringed in yellow, from the side it can be seen from: where it clicks in.

Insert direction

Some parts slide in sideways (axles through bearings, gears onto axles). The picture shows where they end up; push them in along their axle.

Build overview

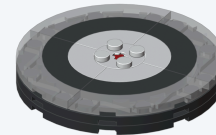
The model is built in 8 sections. Each is finished before it is used.



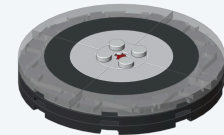
1. Chassis
steps 1–32



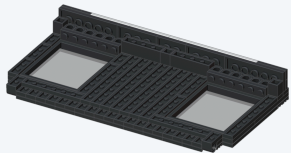
2. VHS Cassette
steps 33, 40, 47–48, 98, 104



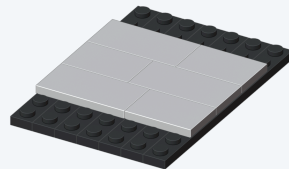
3. Supply reel (full)
steps 34–39



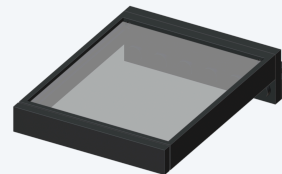
4. Take-up reel
steps 41–46



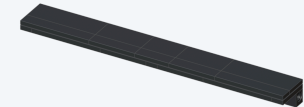
5. Top shell
steps 49–75, 79–90, 96–97



6. Face label
steps 76–78



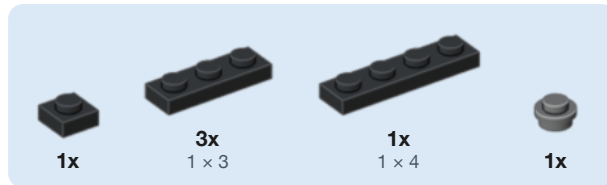
7. Window x2
steps 91–95



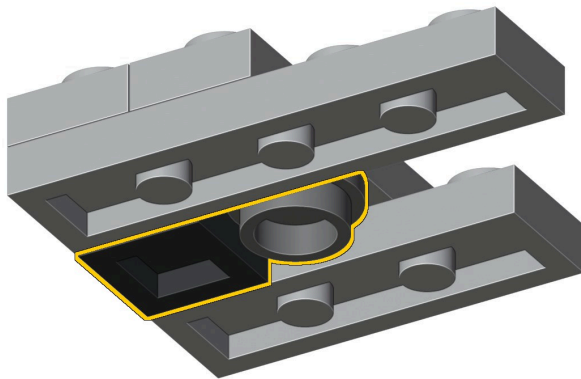
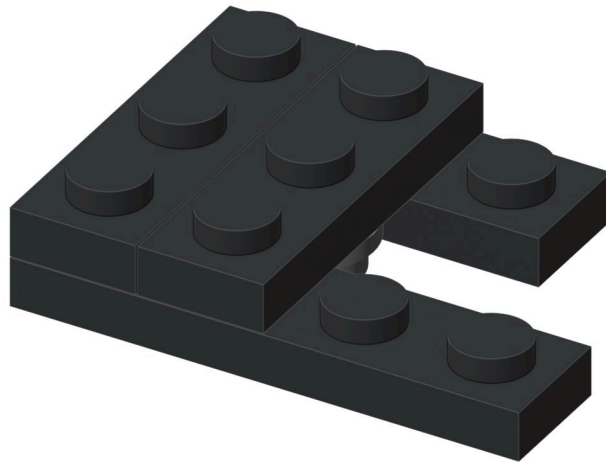
8. Dust flap
steps 99–103

CHASSIS

1



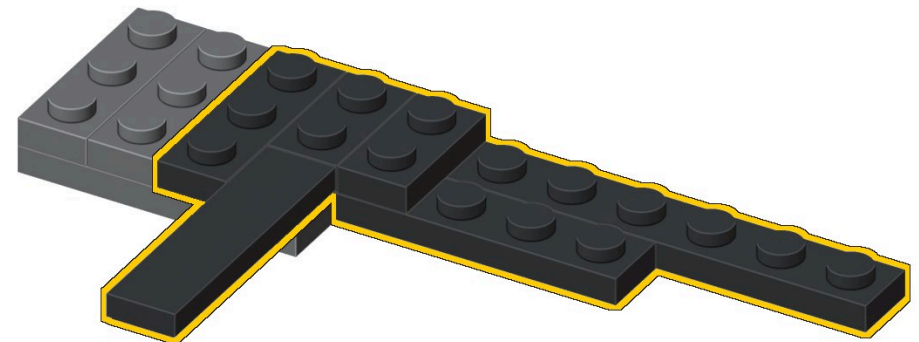
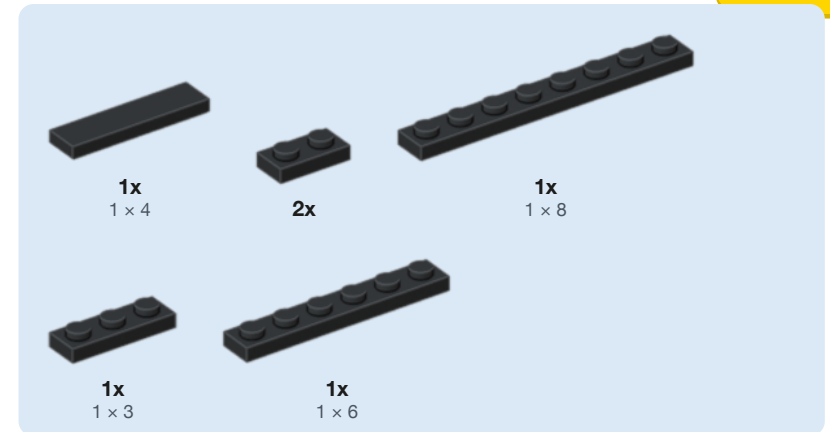
Chassis, bottom layer: the Technic plates' middle holes are the reel drive holes and the lamp hole; silver screw heads sit flush underneath



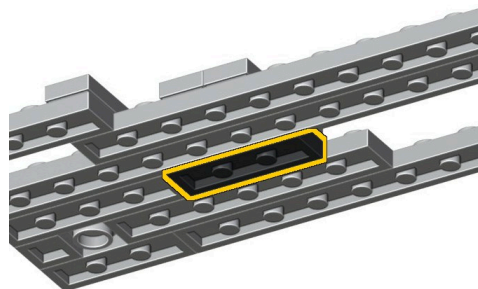
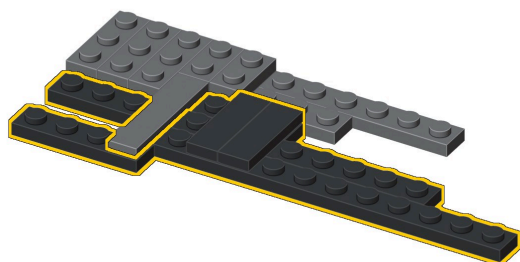
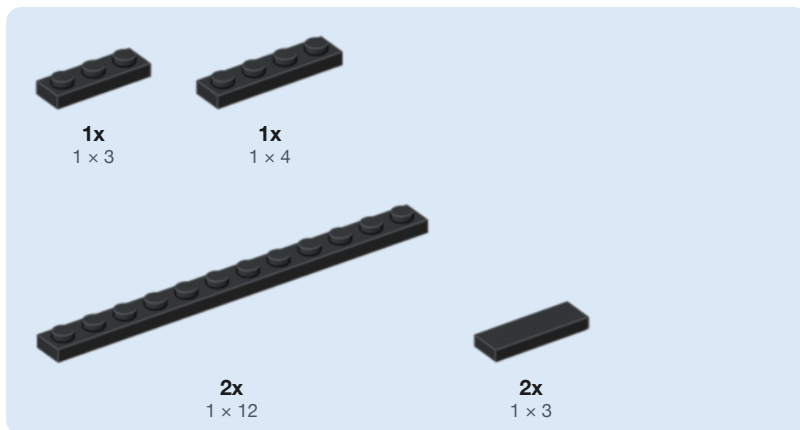
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2

Chassis

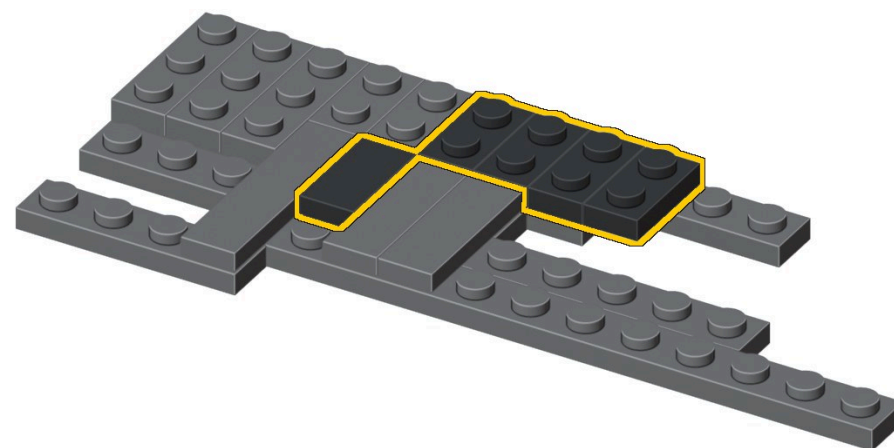
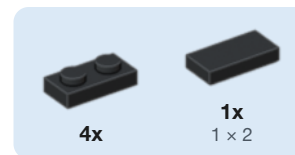


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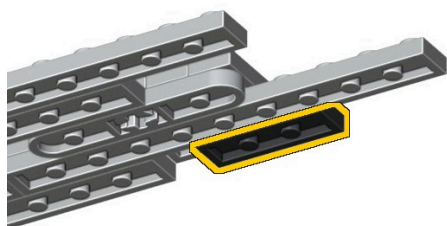
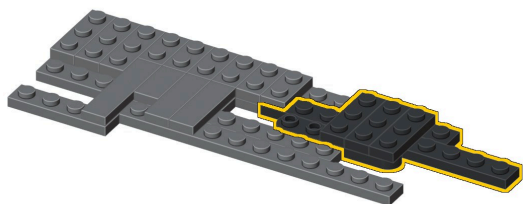
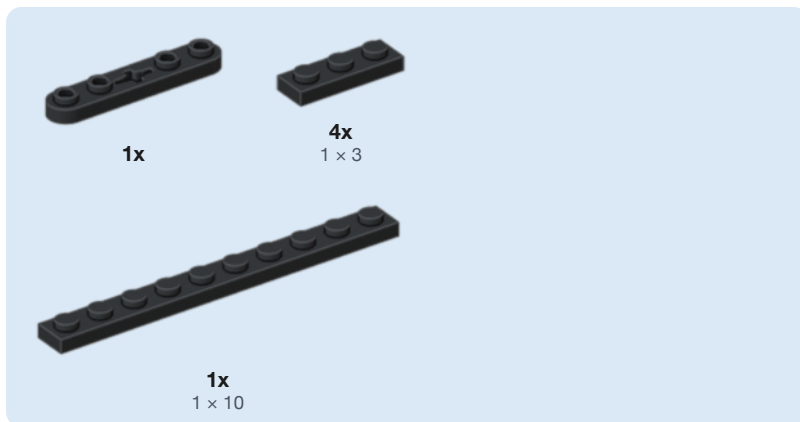
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4



Chassis

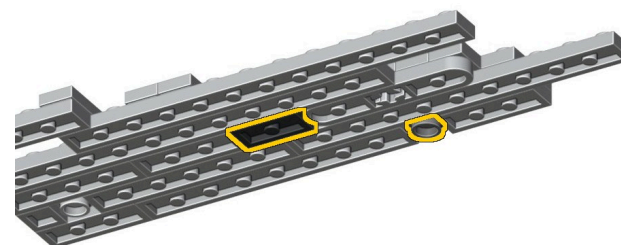
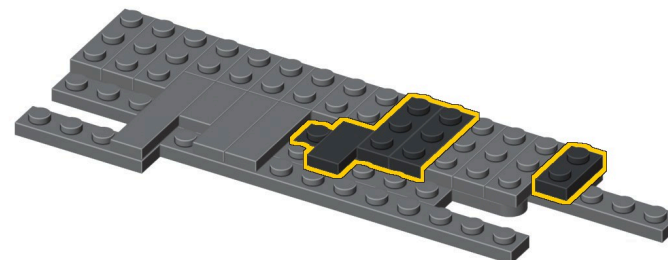
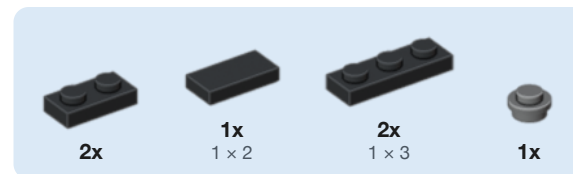
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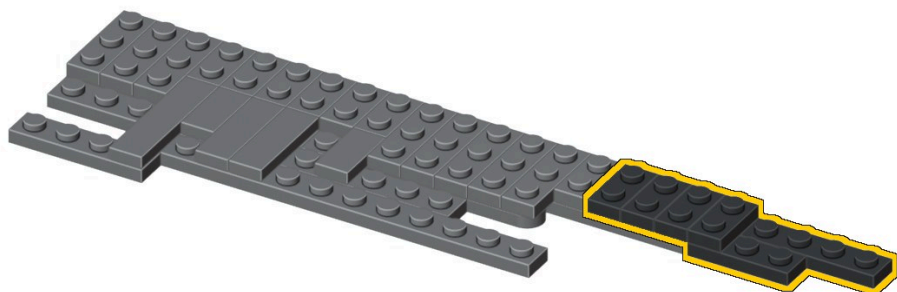
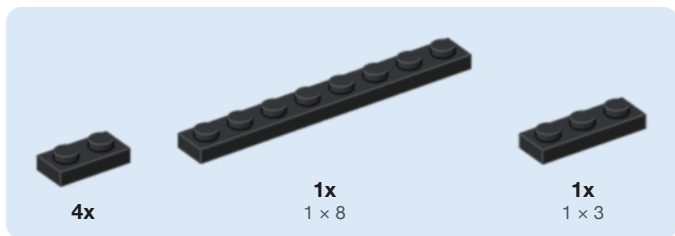
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Chassis



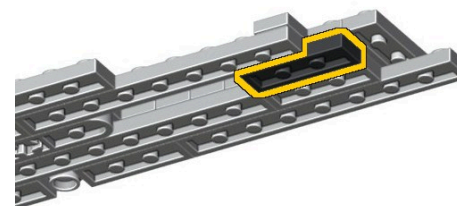
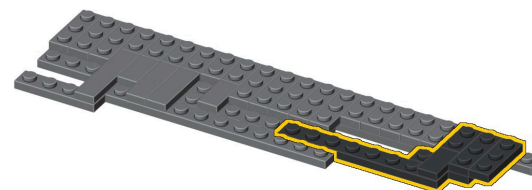
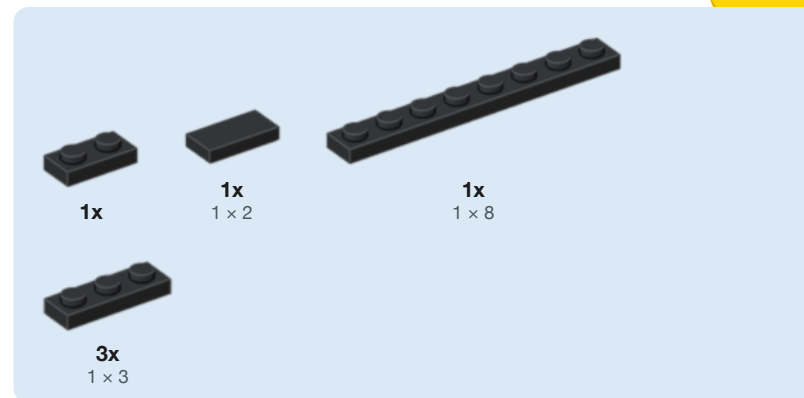
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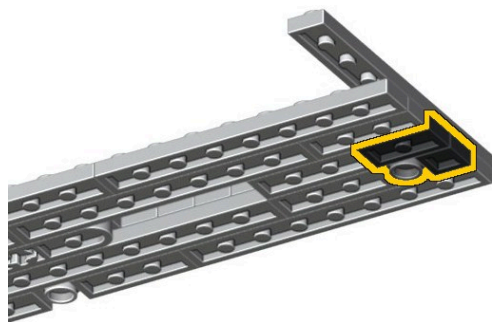
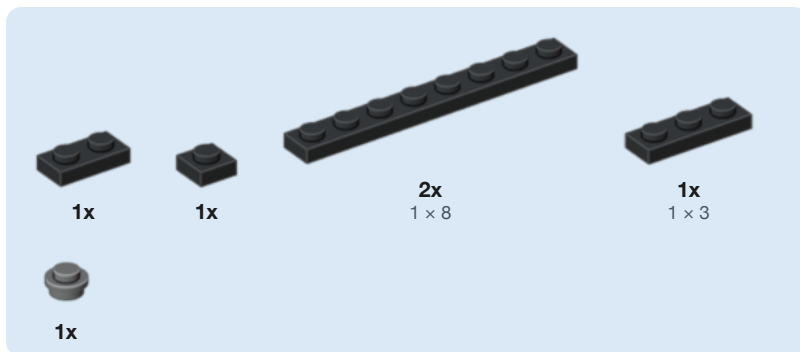
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Chassis



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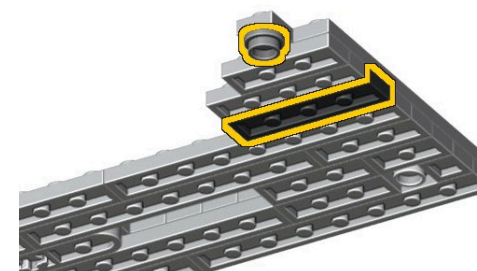
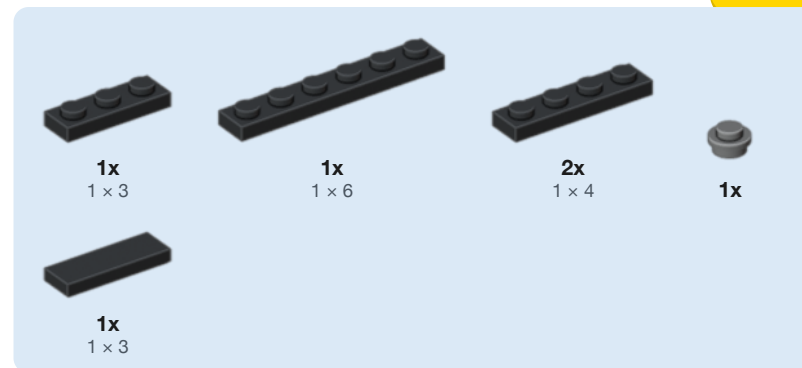
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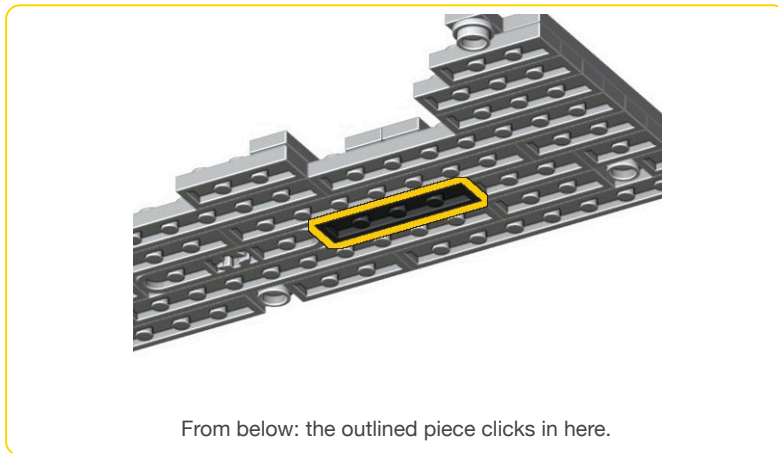
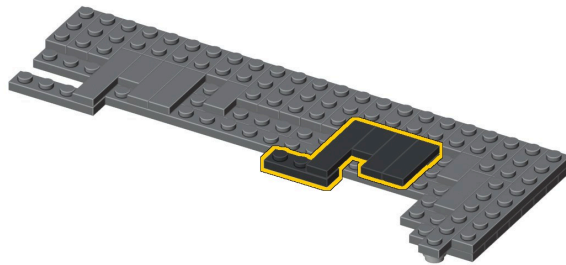
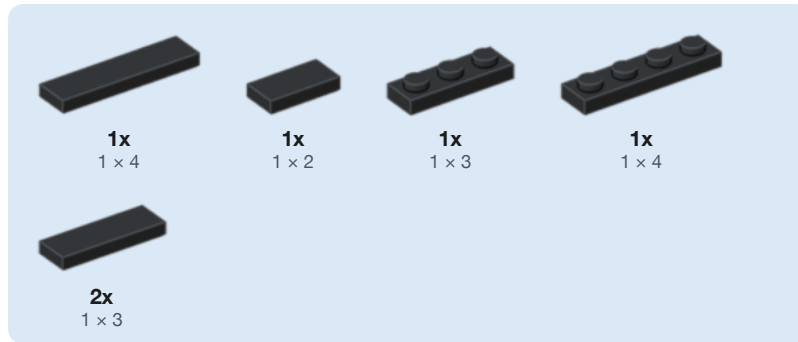
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Chassis



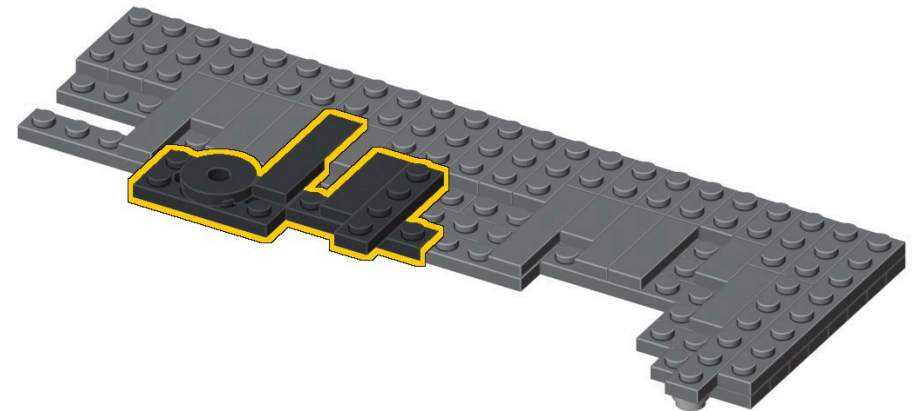
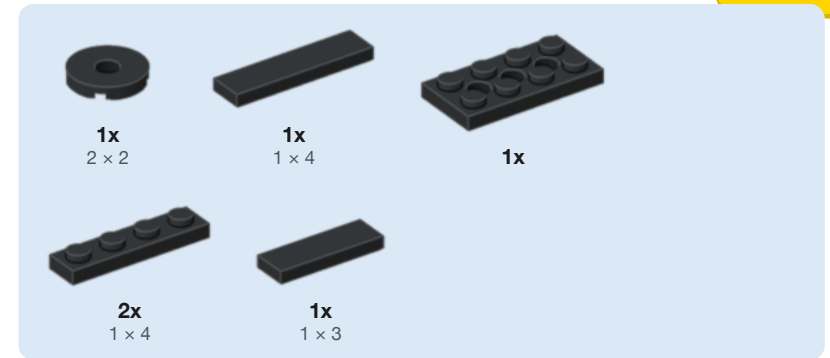
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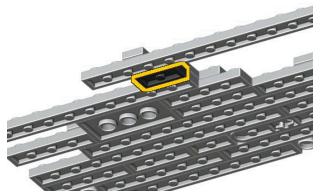
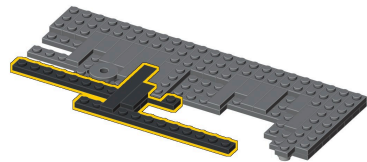
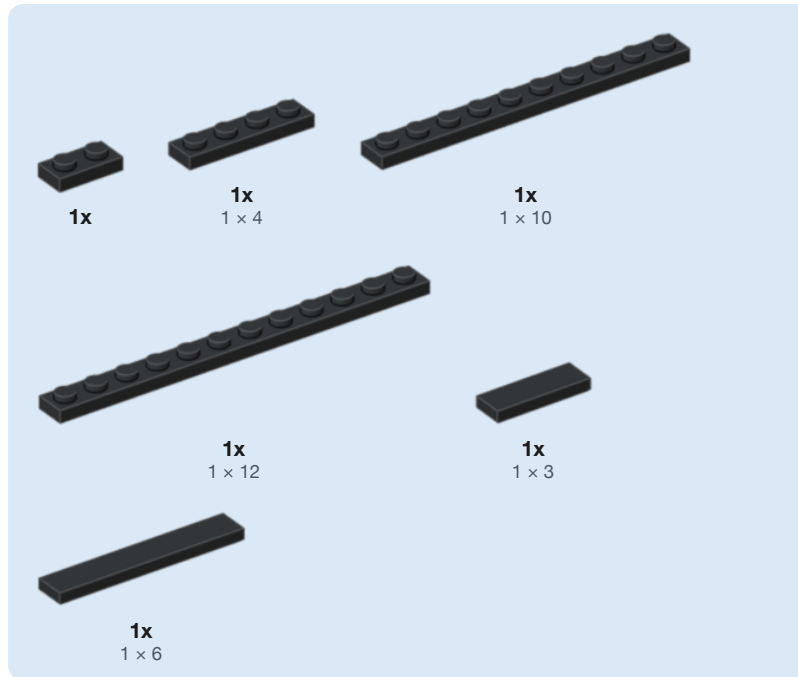


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Chassis

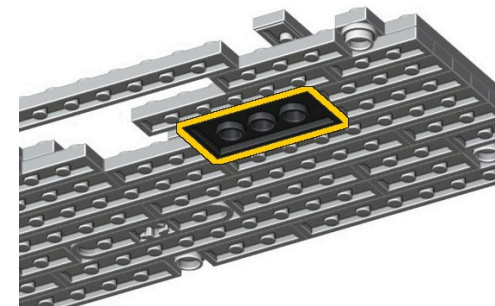
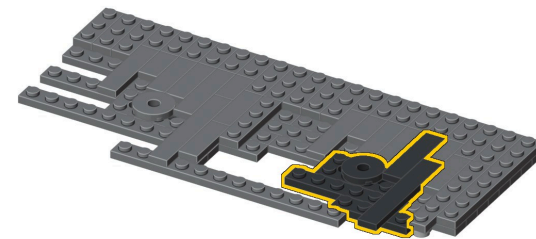
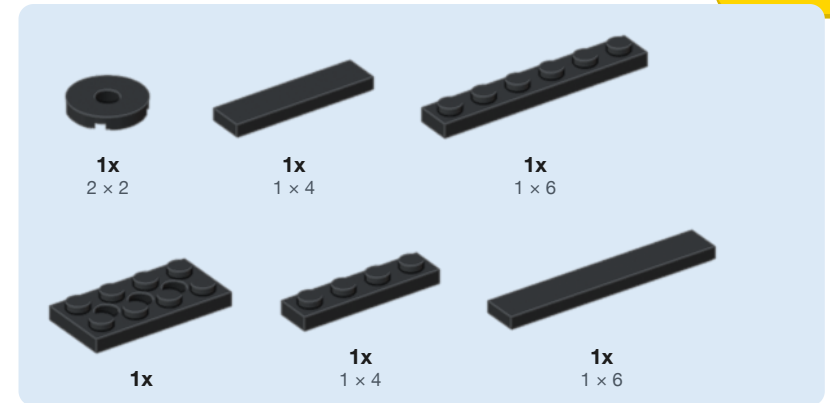


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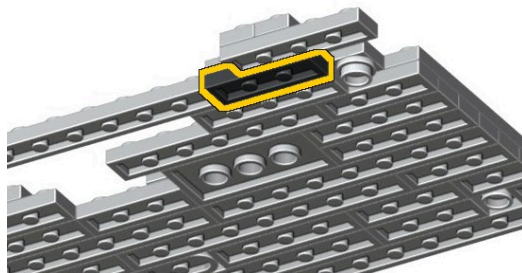
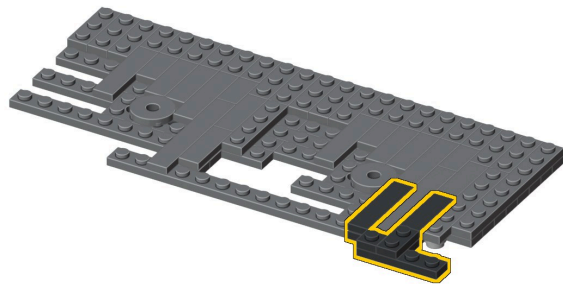
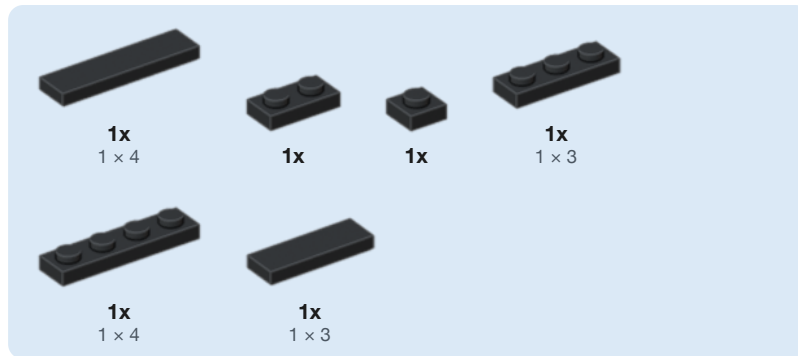
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14



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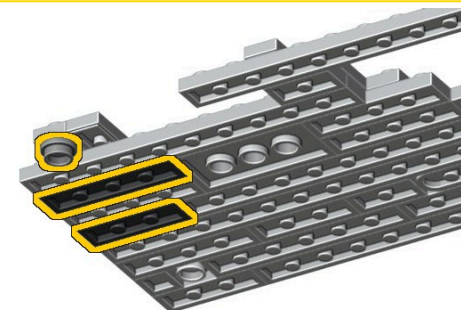
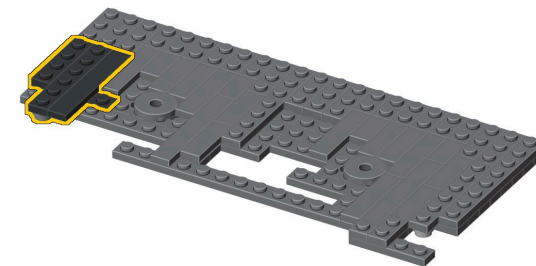
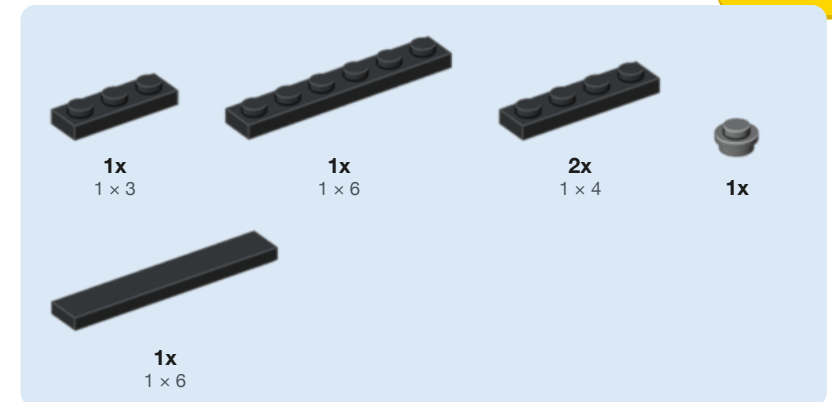
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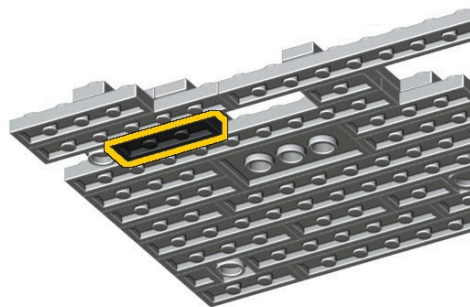
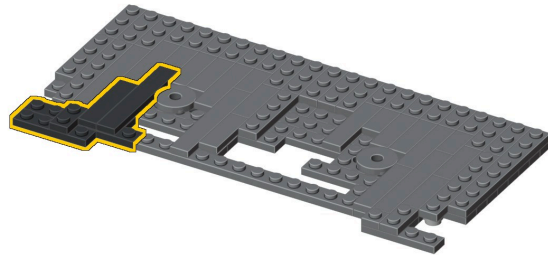
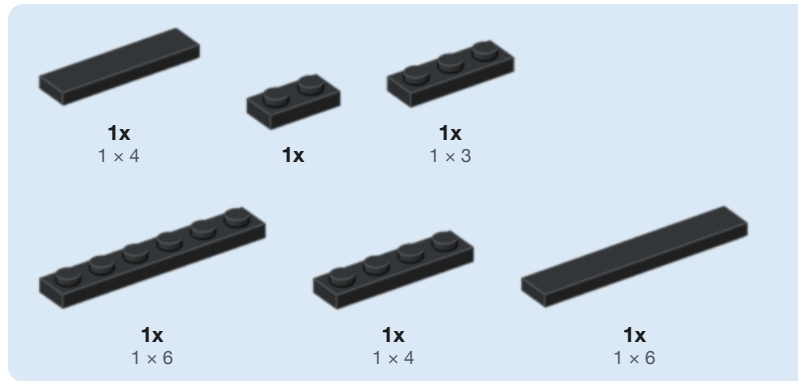
16

Chassis



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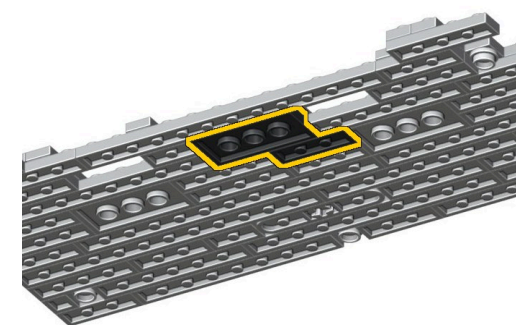
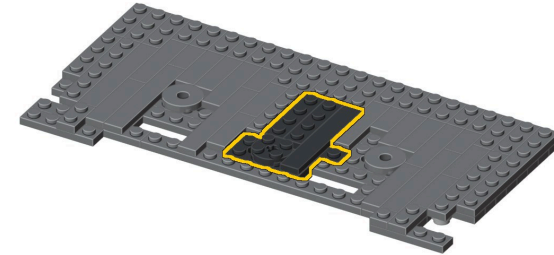
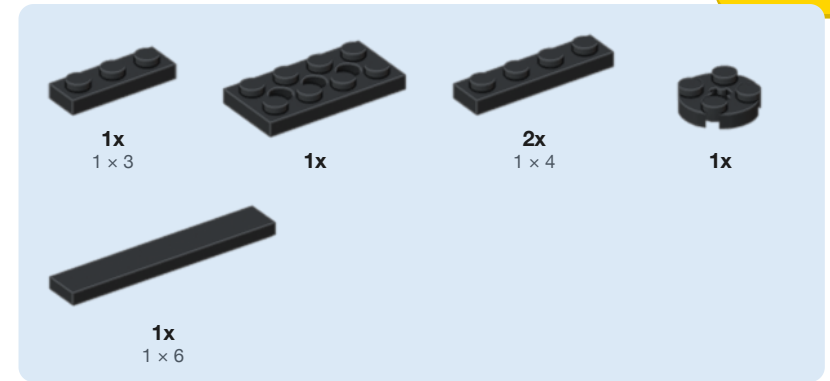
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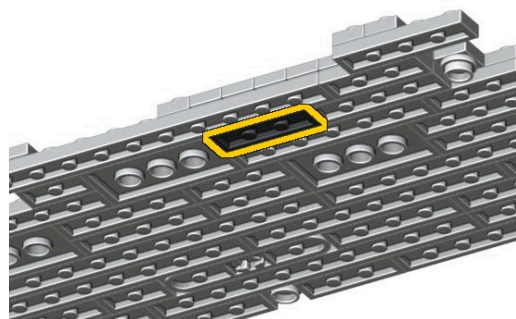
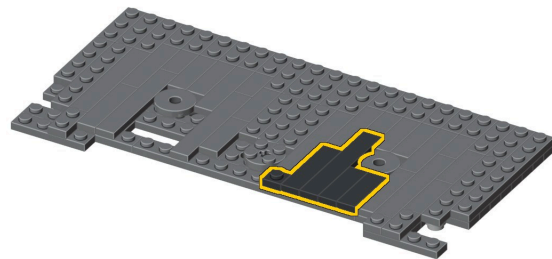
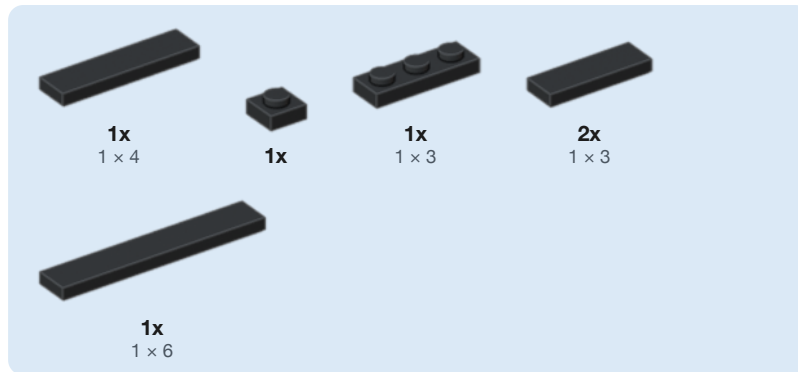
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Chassis



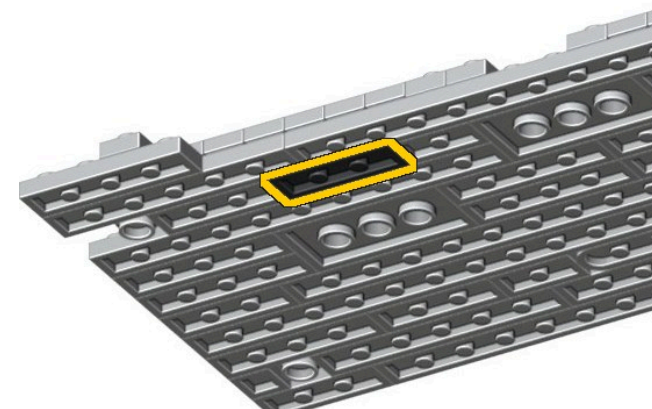
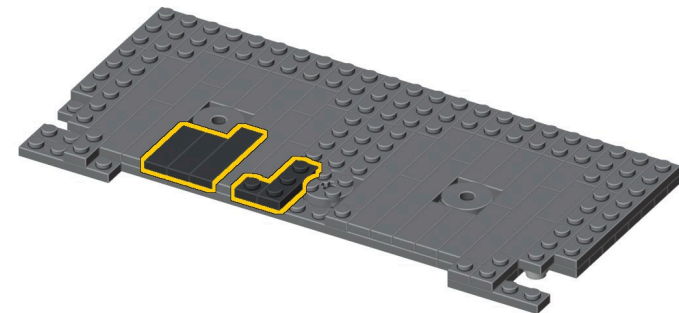
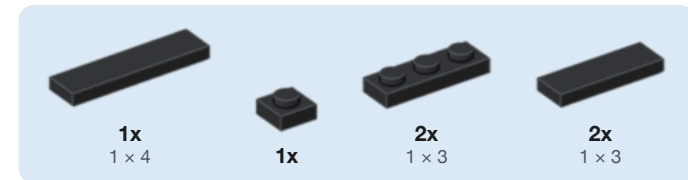
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19



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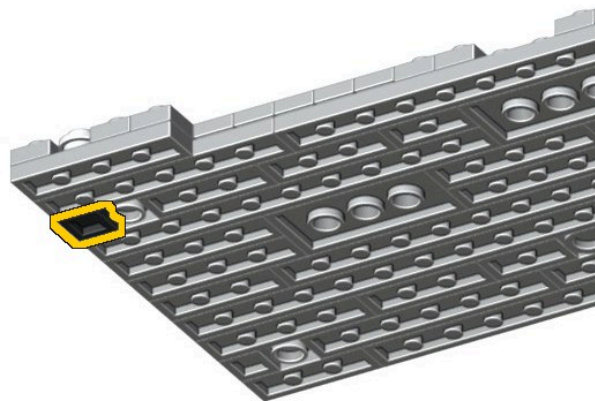
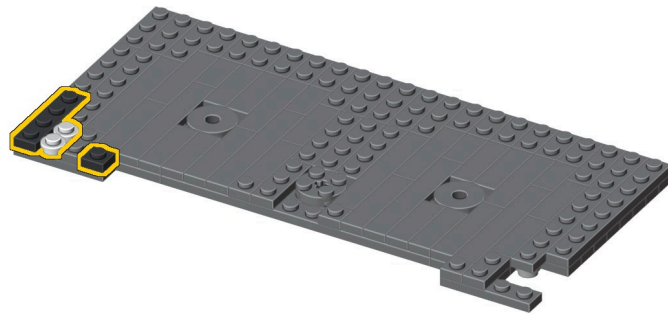
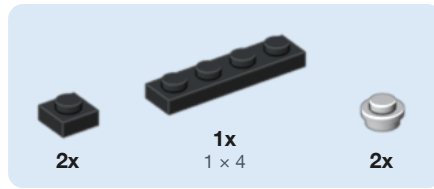
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Chassis

21



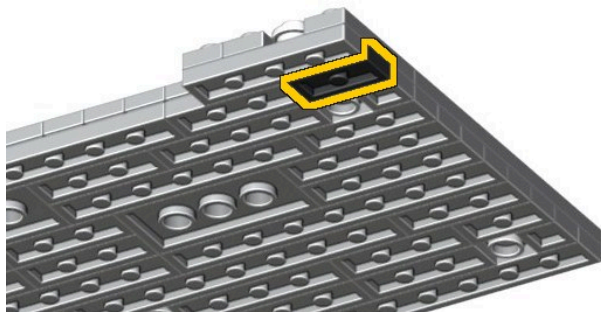
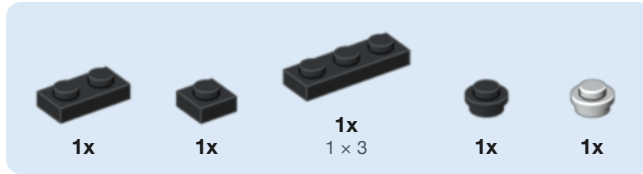
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22



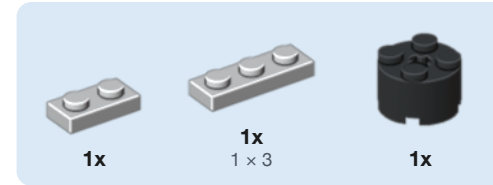
Chassis

23

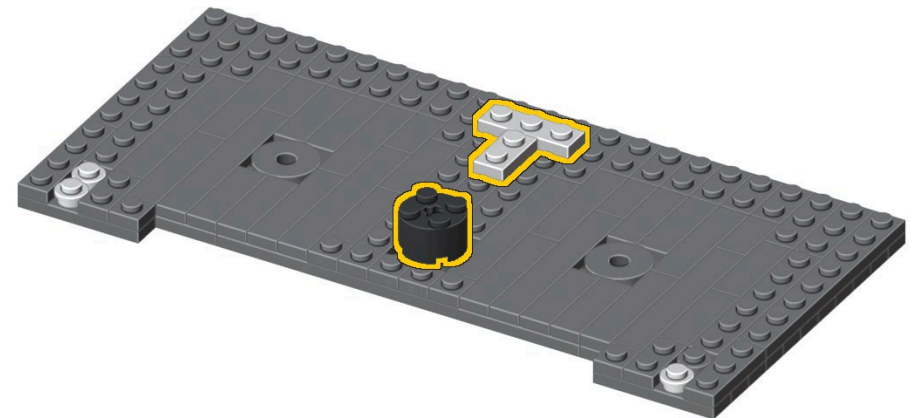


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24

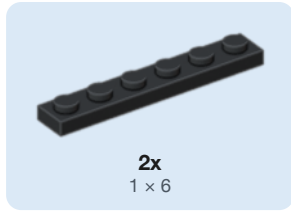


The lamp tube: a black 2 x 2 round brick over the lamp hole

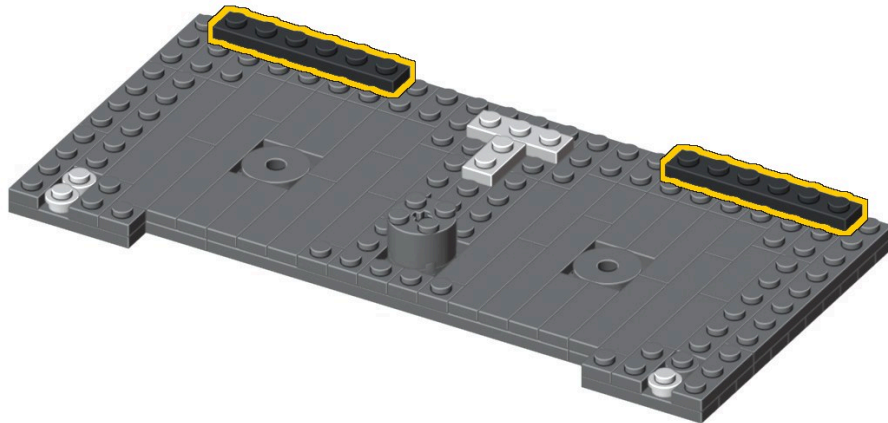


Chassis

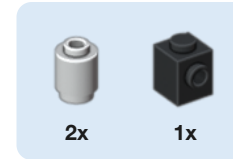
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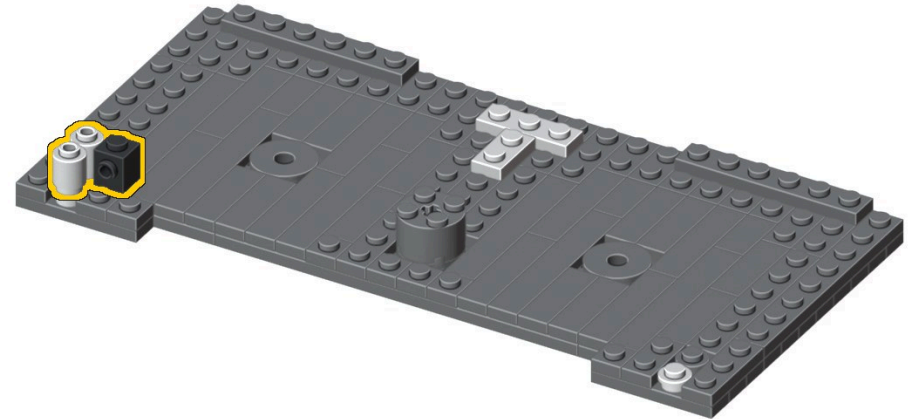
Ledges the window hangers land on



26

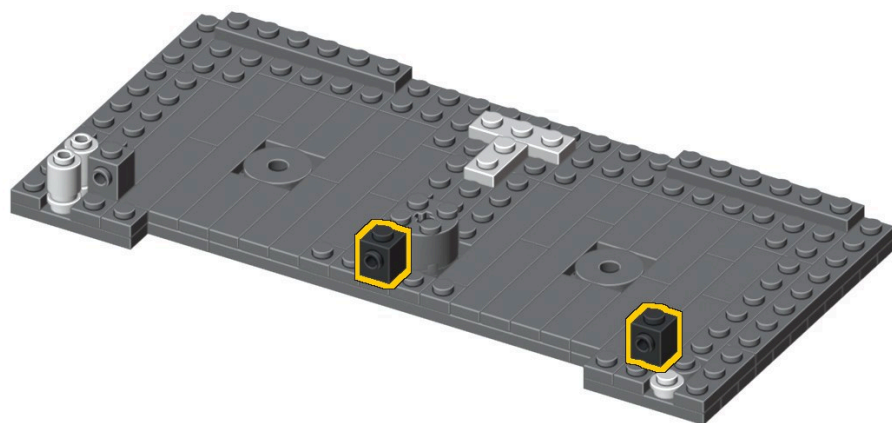


White tape guide rollers at the front corners, a white guide post and the posts the top shell's flap pivots land on

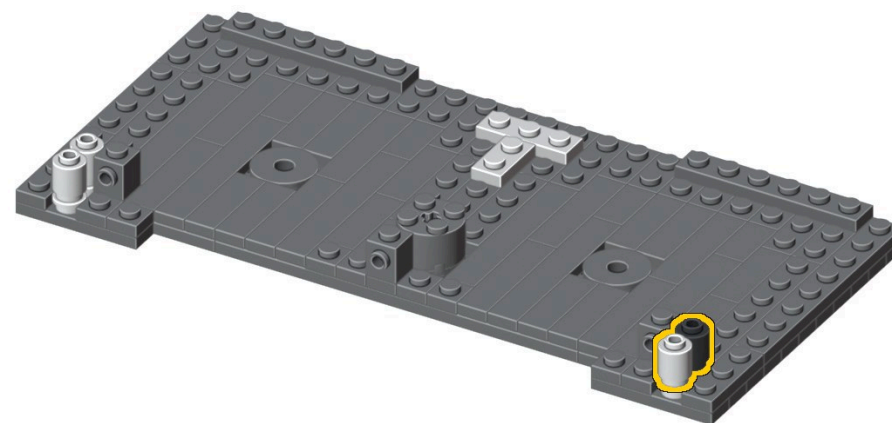
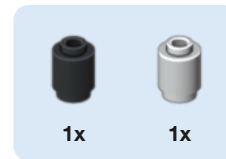


Chassis

27

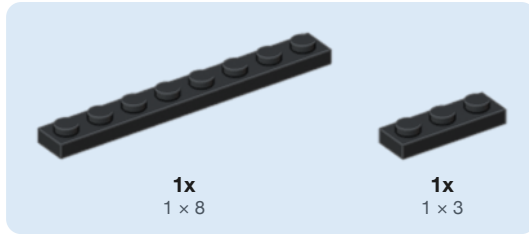


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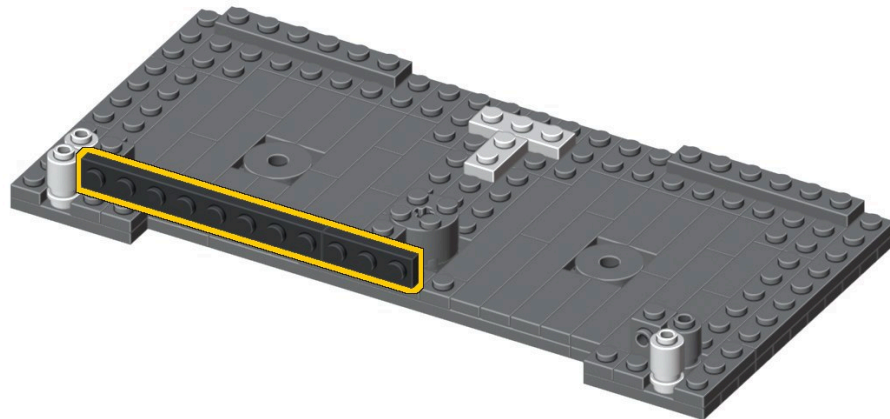


Chassis

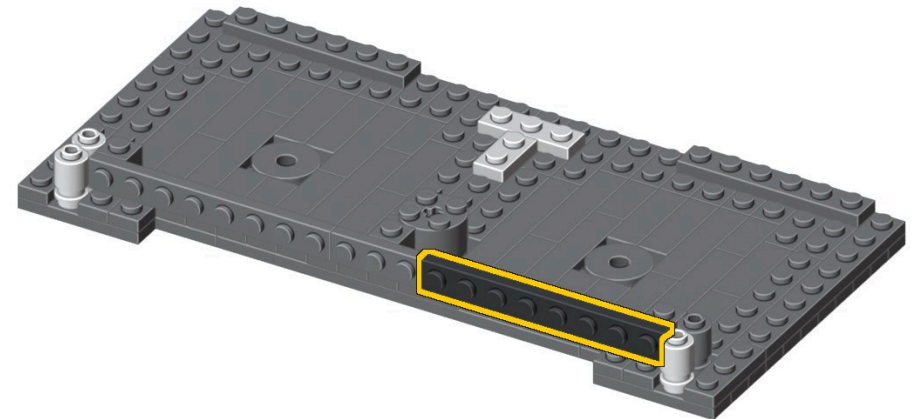
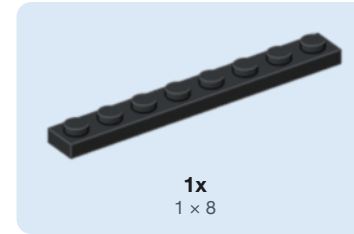
29



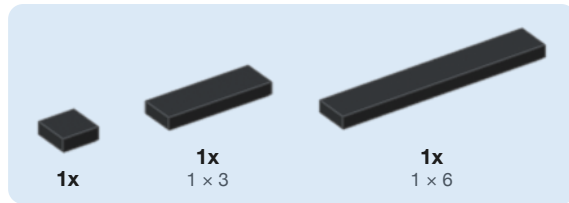
The tape run: a black 1 x N plate stood on its edge



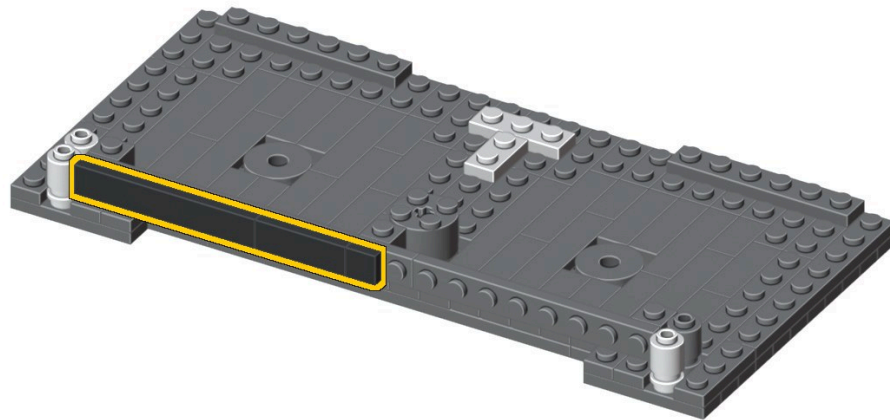
30



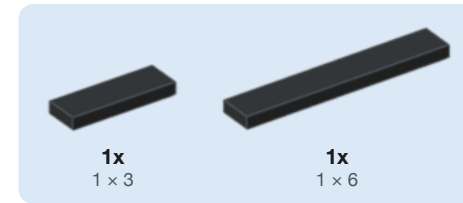
31



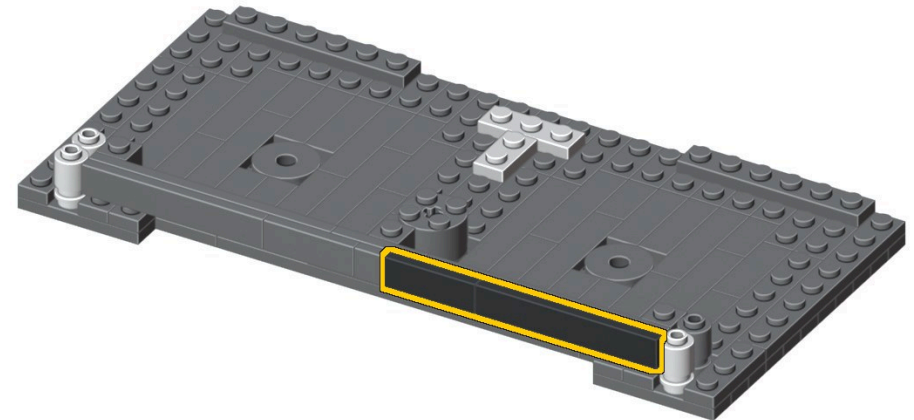
Black tiles on it: the tape stretched across the front



32

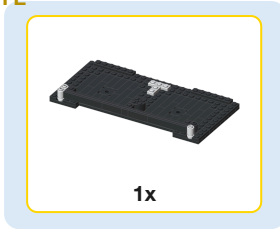


Chassis



VHS CASSETTE

33



The chassis



SUPPLY REEL (FULL)

34



Lower flange: a black 8 x 8 round plate with a hole in the middle

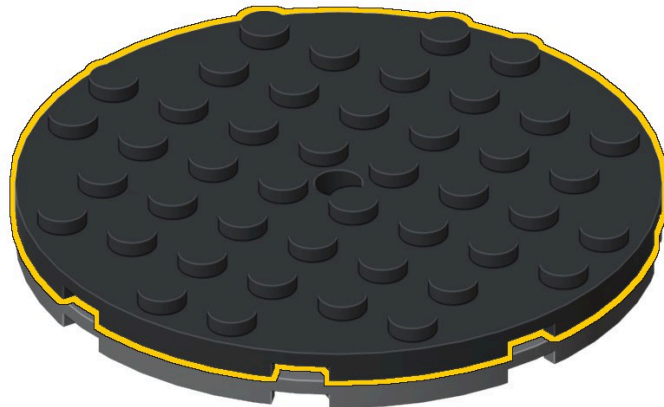


Supply reel (full)

35



The tape pack: a second black 8 x 8 round plate on top of it

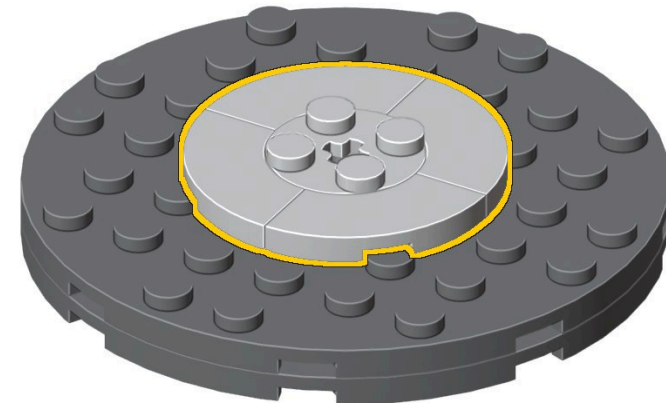


36

Supply reel (full)



The hub: a round plate with an axle hole, and a flat white ring round it



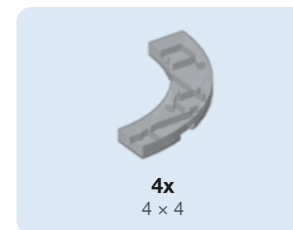
37



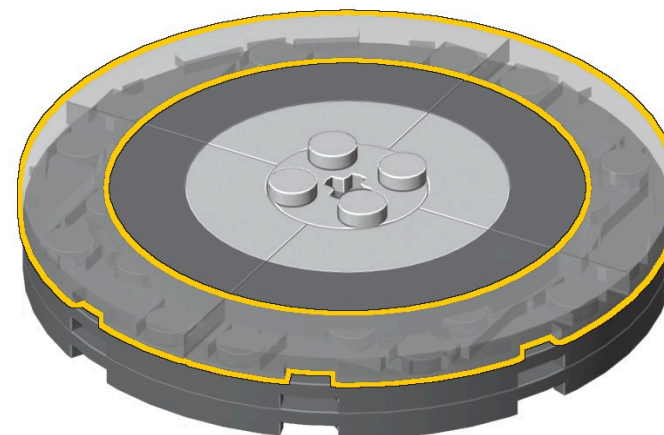
Tape wound round the hub



38

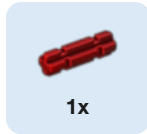


Clear upper flange



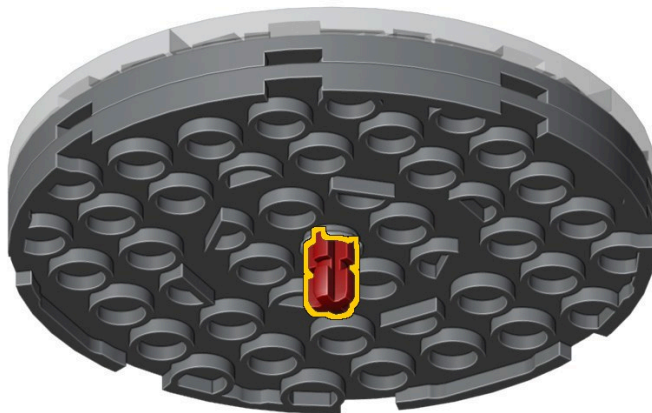
Supply reel (full)

39



seen from below

The red spindle: push the axle up through the hub



VHS CASSETTE

40

VHS Cassette



Drop the full supply reel onto the right-hand bearing: its red axle goes down through the round holes



41



1x
8 x 8

Lower flange: a black 8 x 8 round plate with a hole in the middle

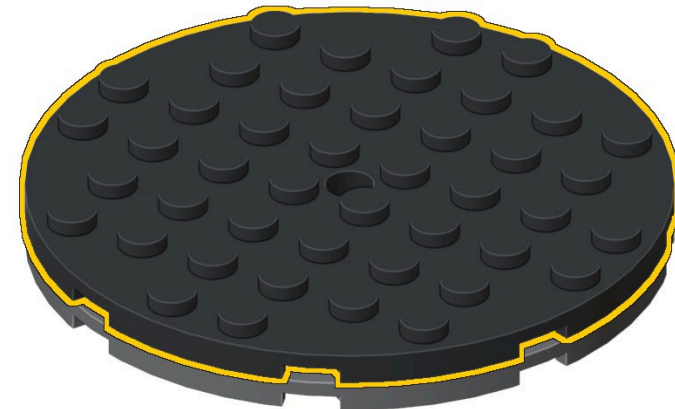


42

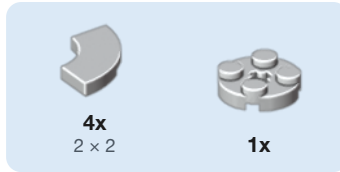


1x
8 x 8

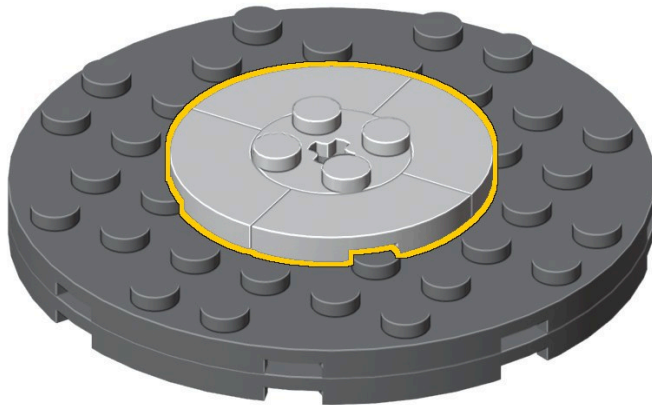
The tape pack: a second black 8 x 8 round plate on top of it



43



The hub: a round plate with an axle hole, and a flat white ring round it



44



Just a few turns of tape left on the hub

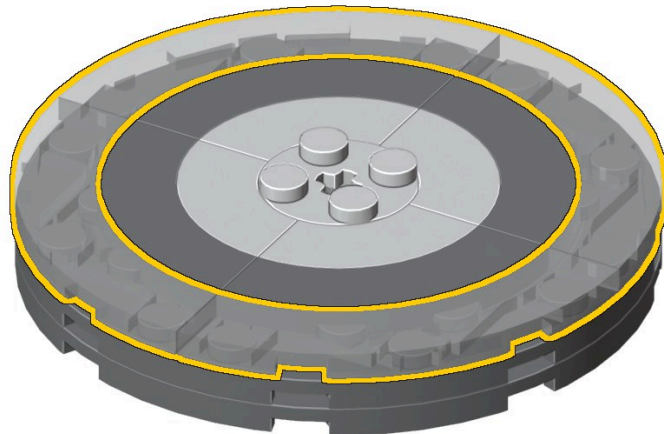


Take-up reel

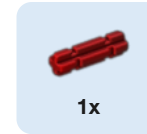
45



Clear upper flange



46



The red spindle: push the axle up through the hub

Take-up reel

seen from below



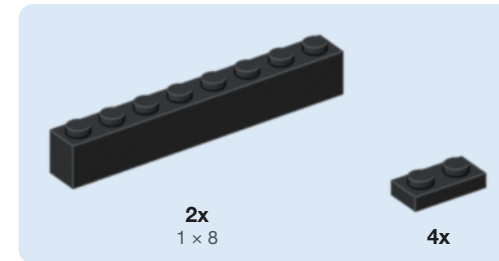
47



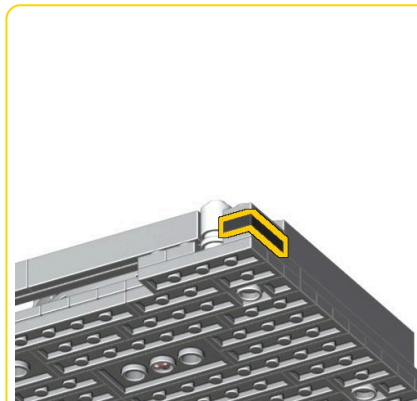
The take-up reel on the left-hand bearing



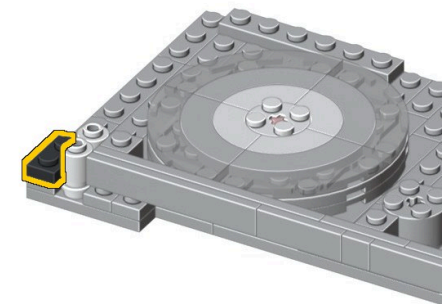
48



End walls



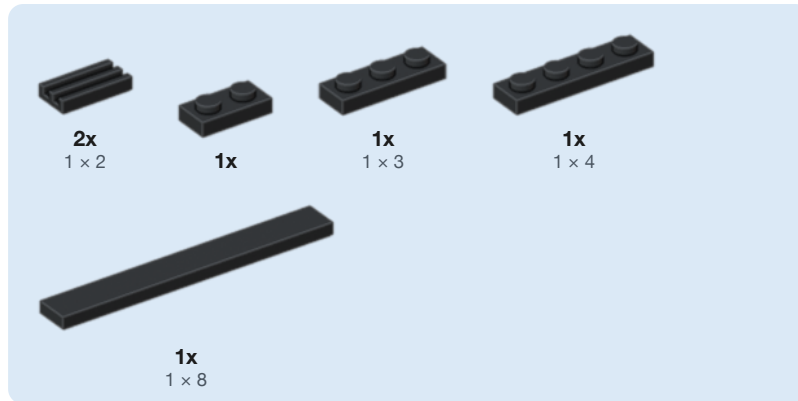
From below: the outlined piece clicks in here.



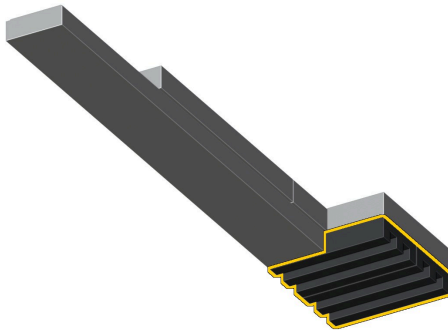
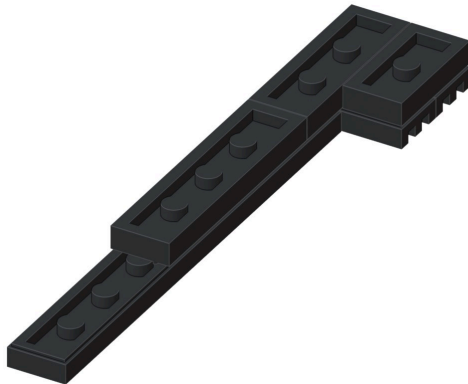
First the outlined piece: the others go on over it.

TOP SHELL

49

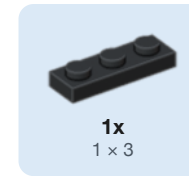


Top shell, built upside down: the grip-band grilles and the black rims go face down, then plates on them

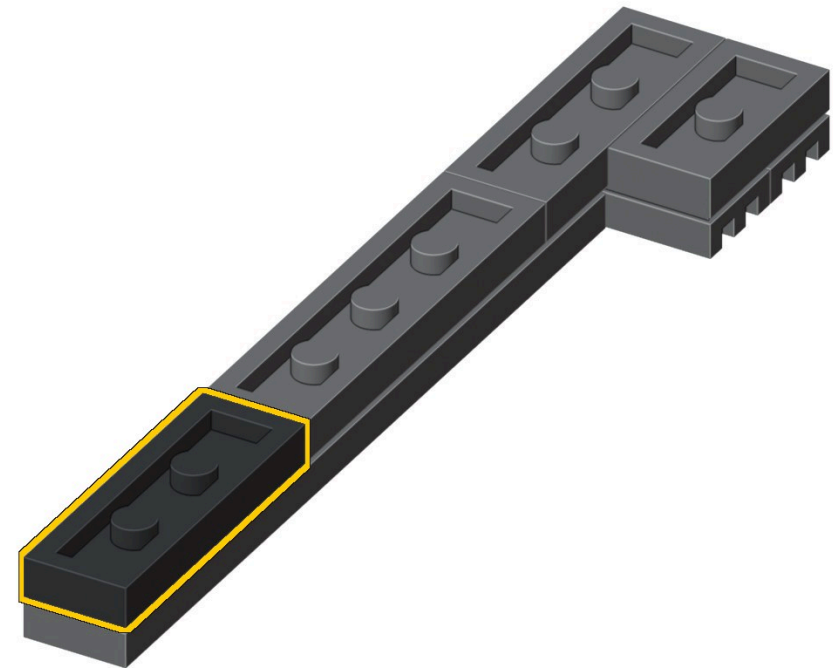


From below: the outlined pieces click in here.

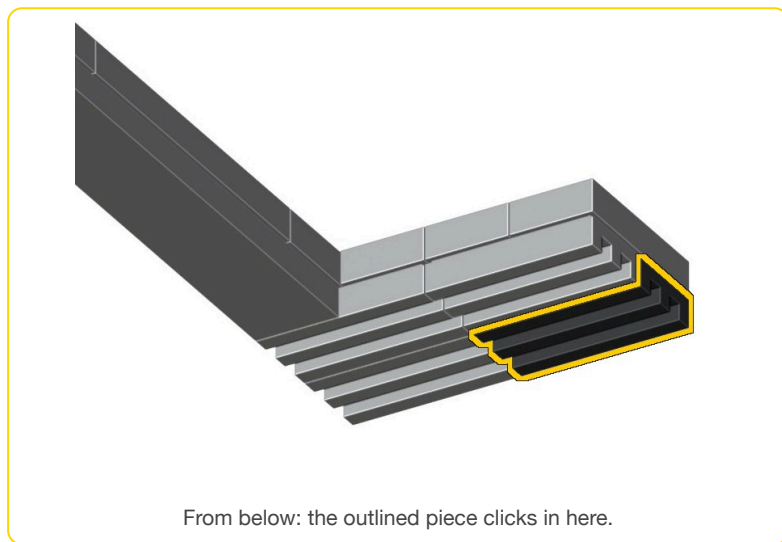
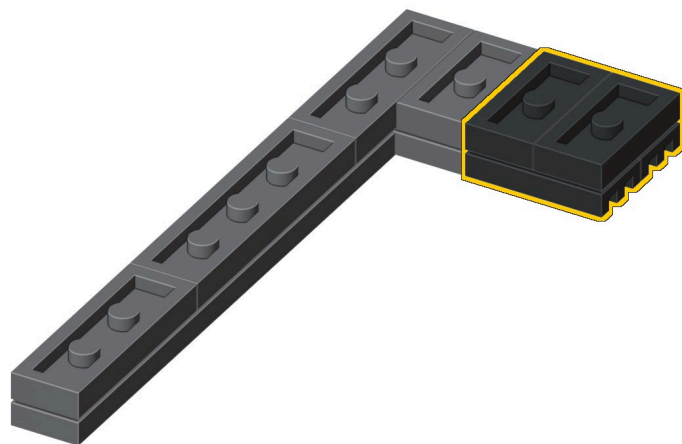
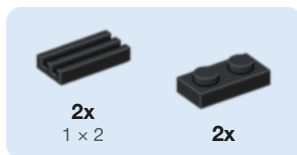
50



Top shell

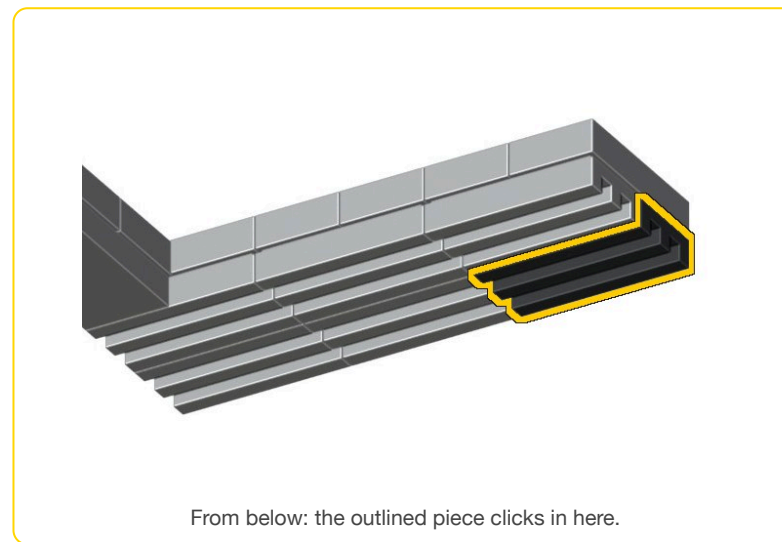
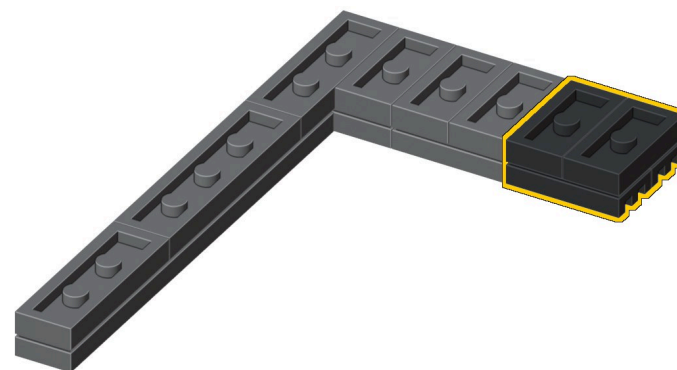
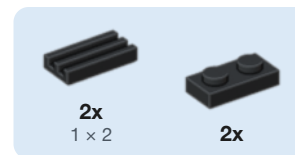


51

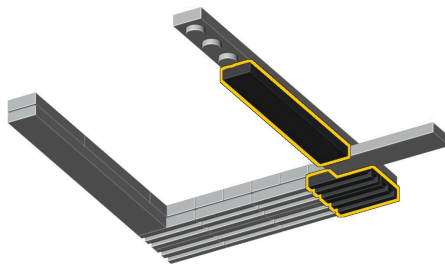
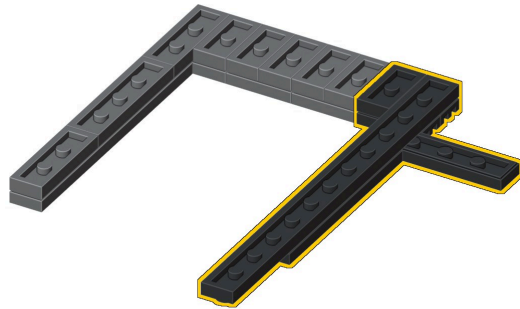
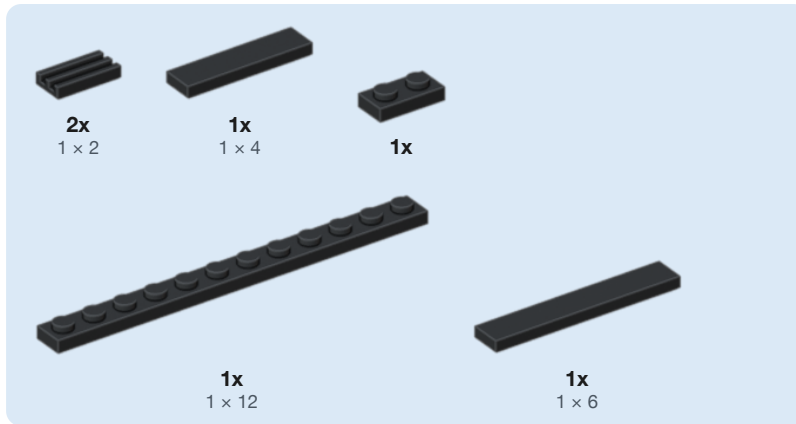


52

Top shell

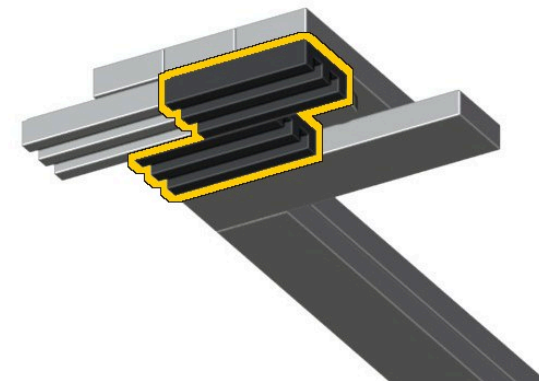
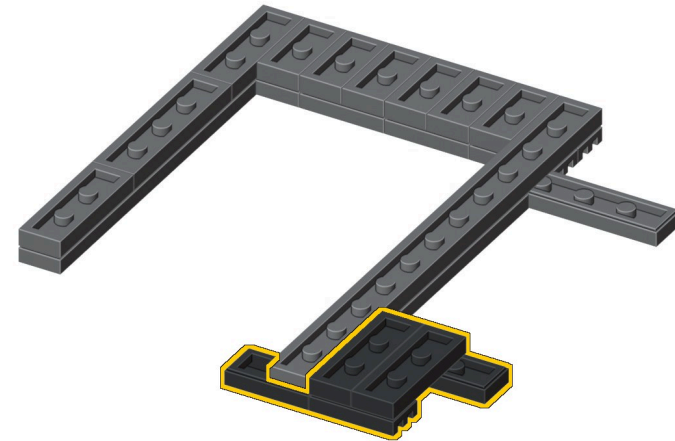
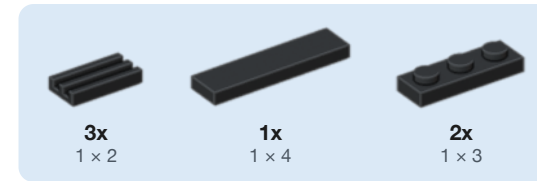


53



From below: the outlined pieces click in here.

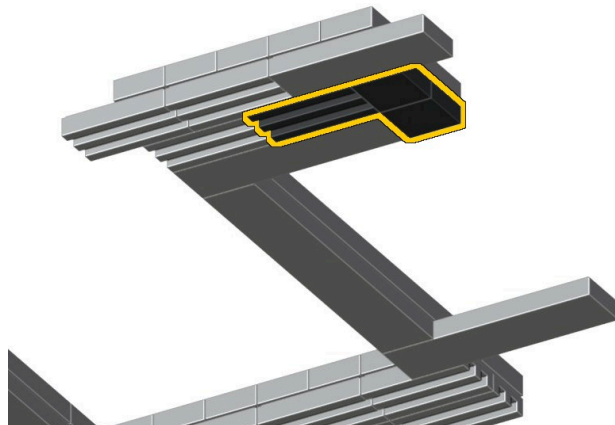
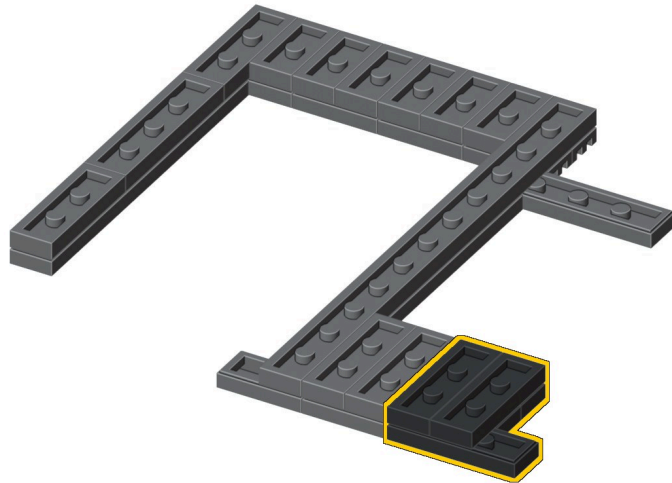
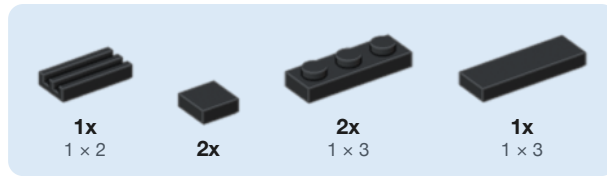
54



From below: the outlined pieces click in here.

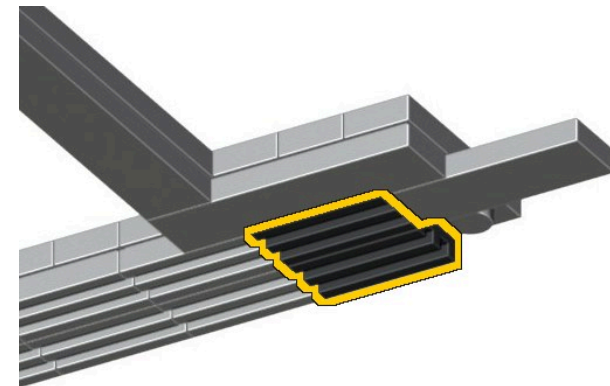
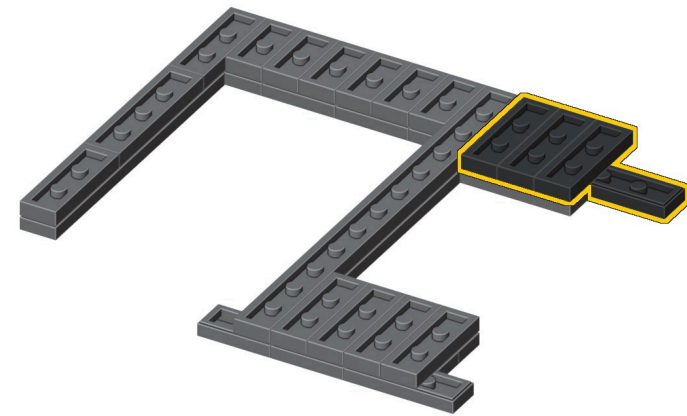
Top shell

55



From below: the outlined pieces click in here.

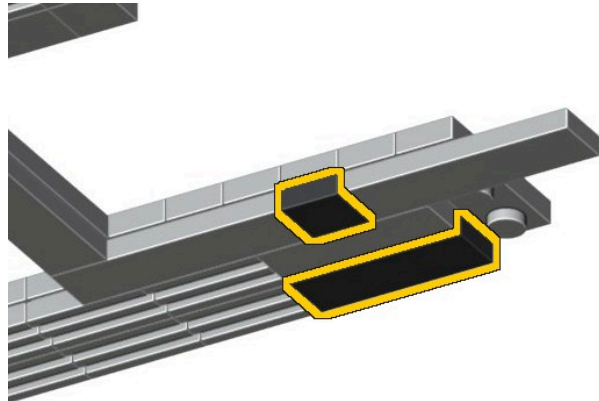
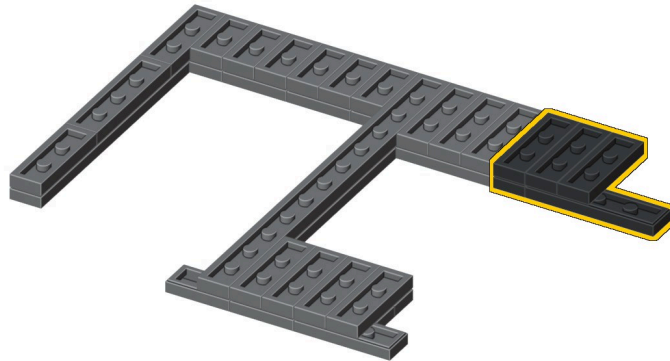
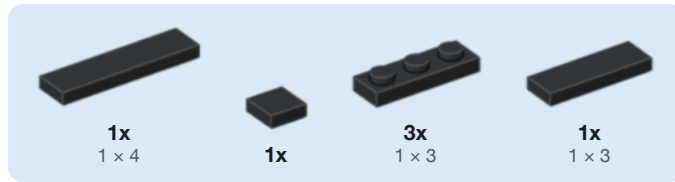
56



From below: the outlined pieces click in here.

Top shell

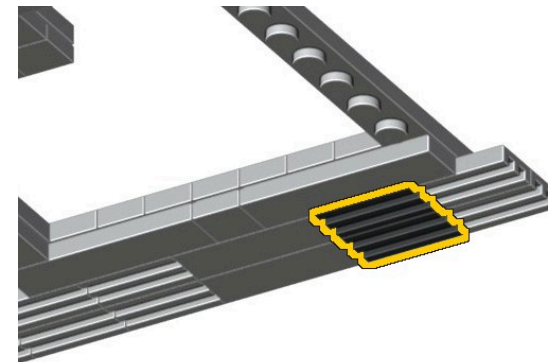
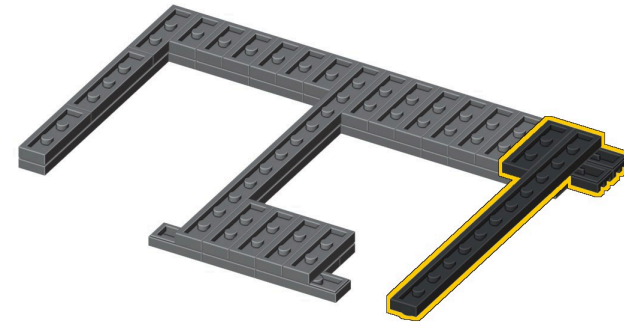
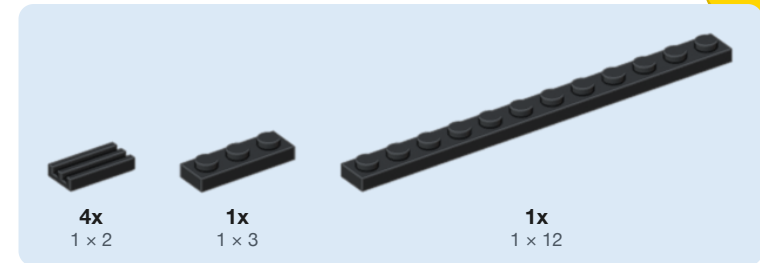
57



From below: the outlined pieces click in here.

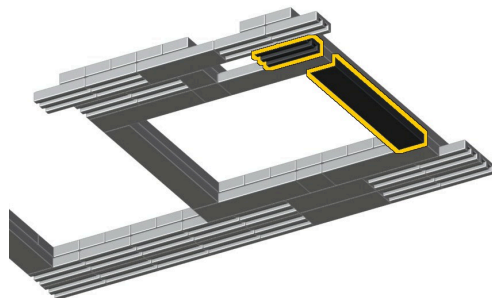
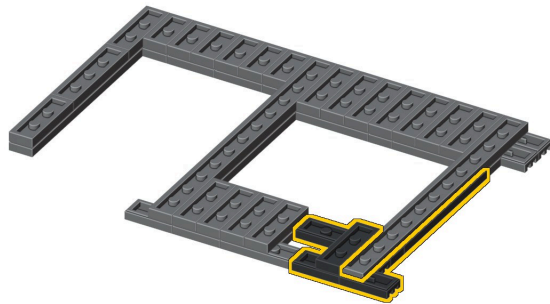
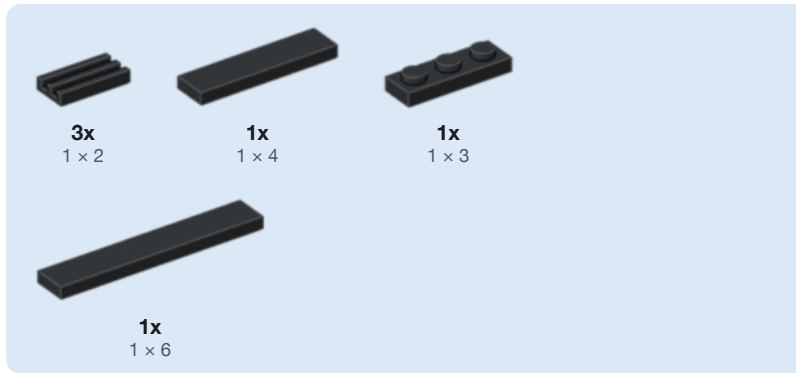
58

Top shell



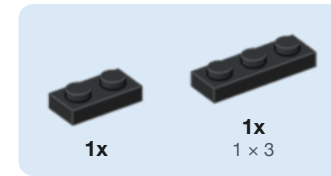
From below: the outlined pieces click in here.

59



From below: the outlined pieces click in here.

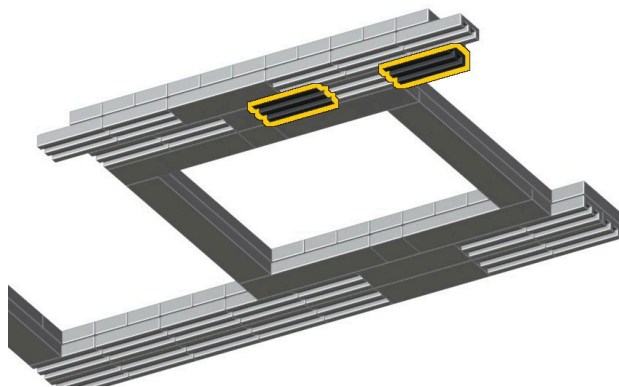
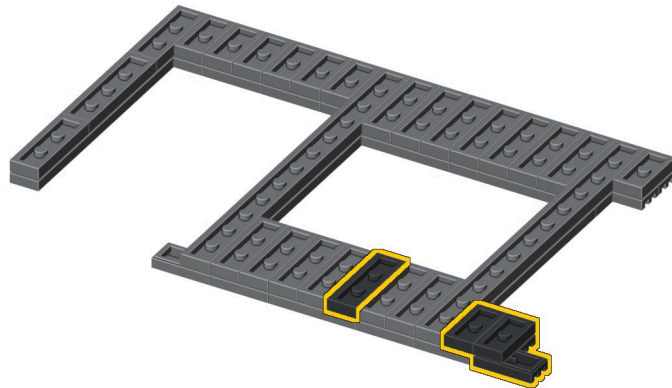
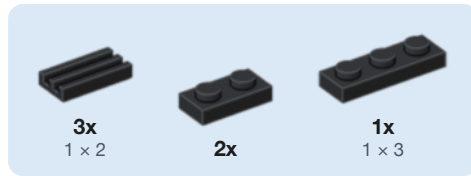
60



Top shell

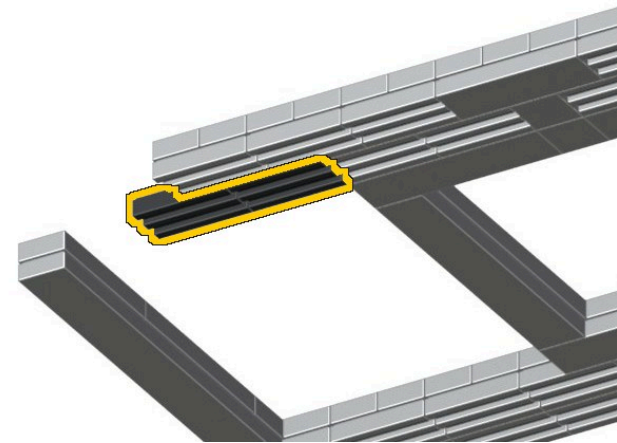
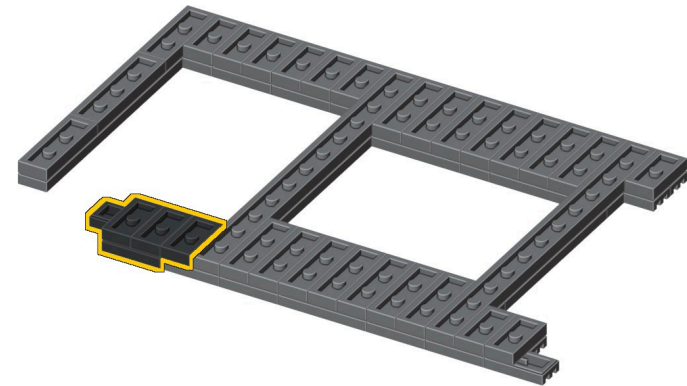
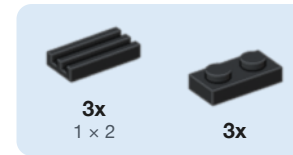


61



From below: the outlined pieces click in here.

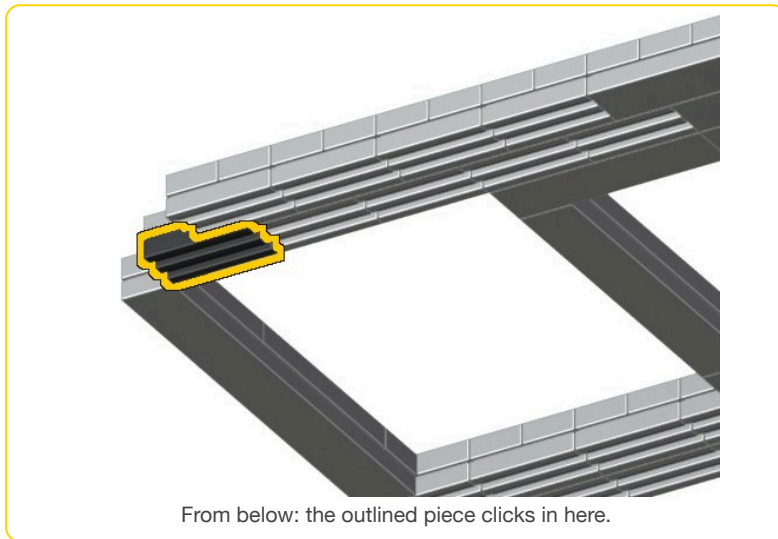
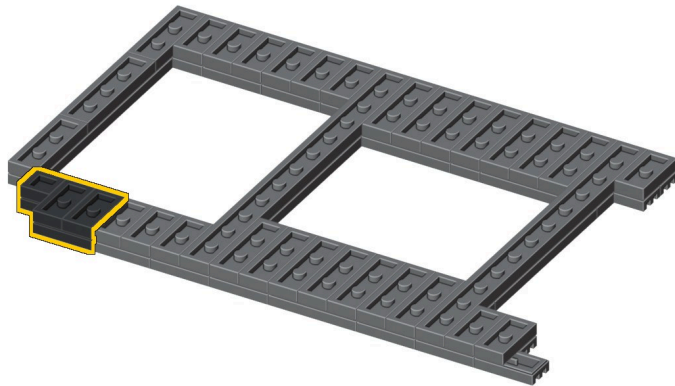
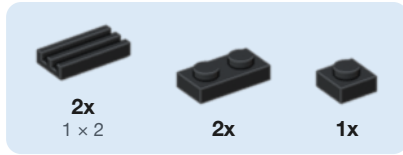
62



From below: the outlined pieces click in here.

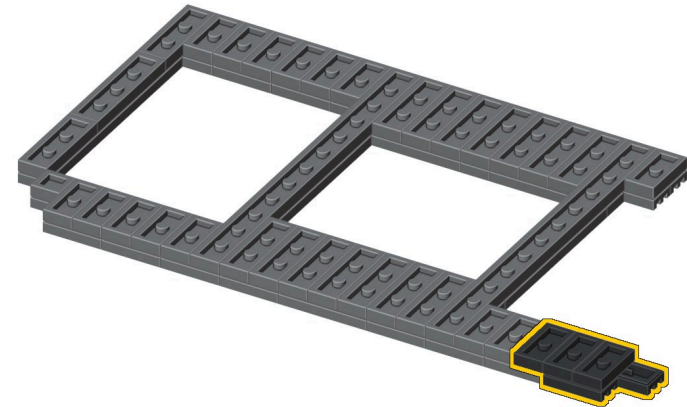
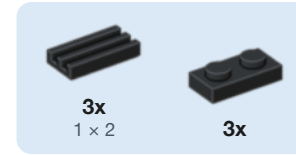
Top shell

63

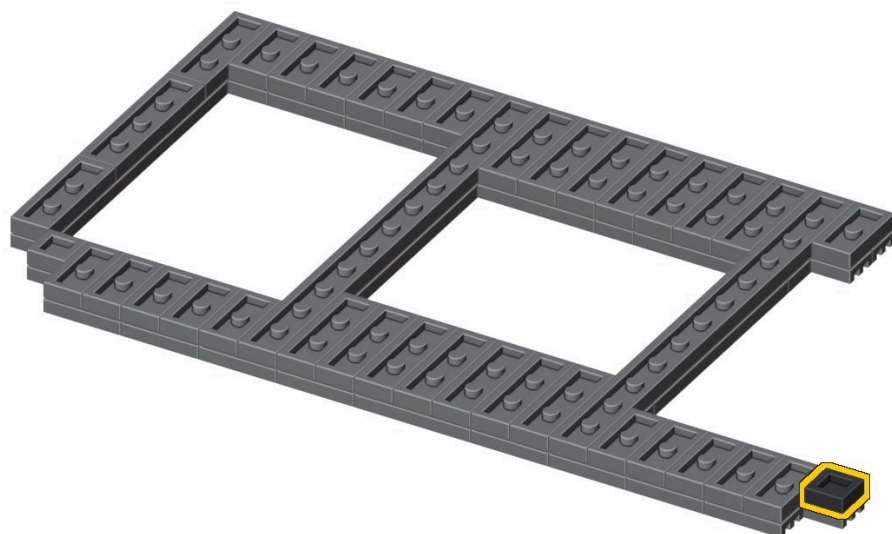
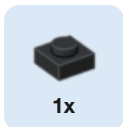


64

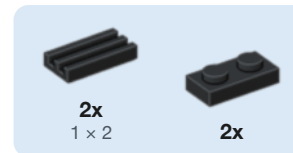
Top shell



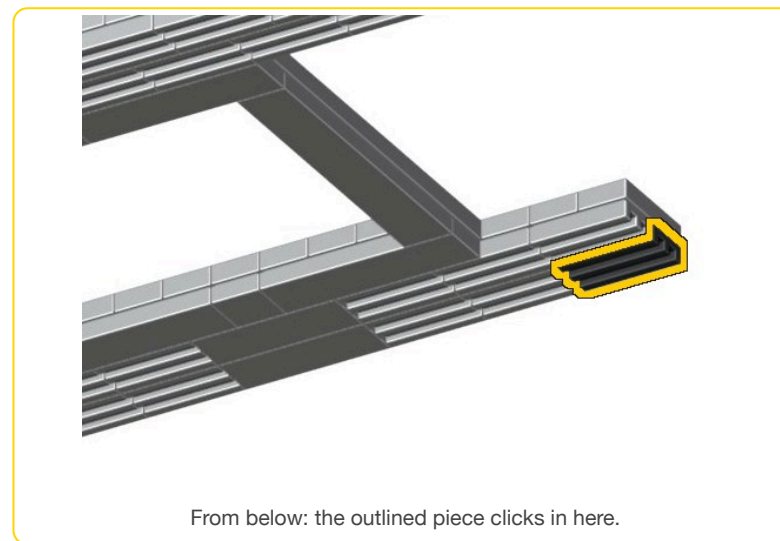
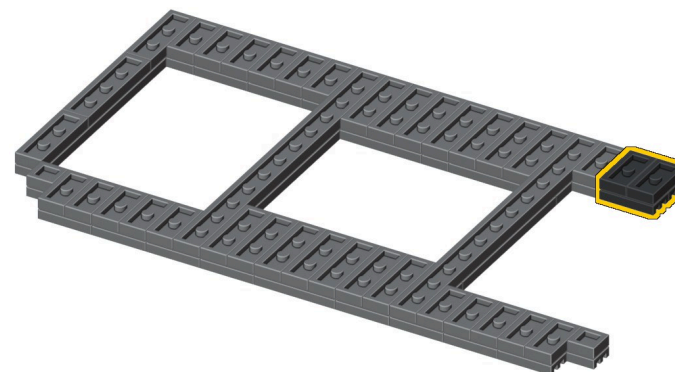
65



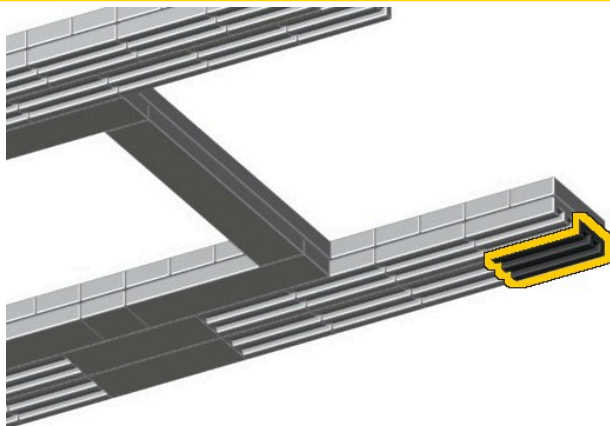
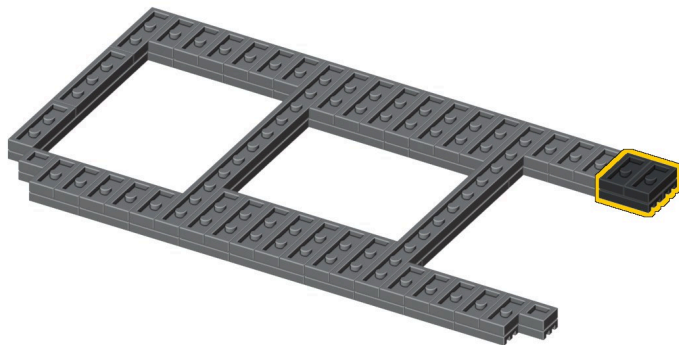
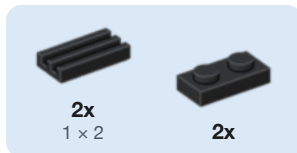
66



Top shell



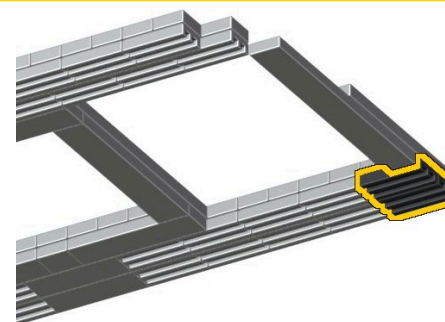
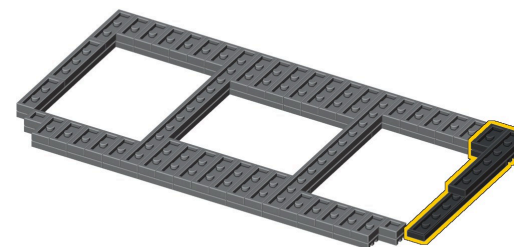
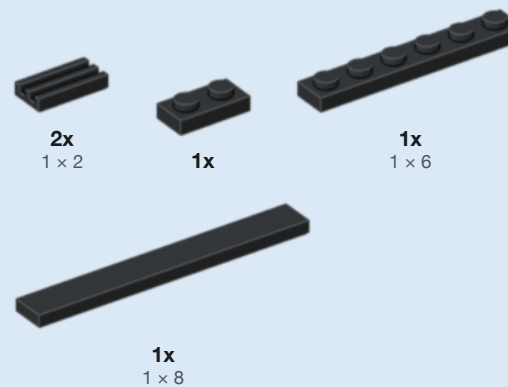
67



From below: the outlined piece clicks in here.

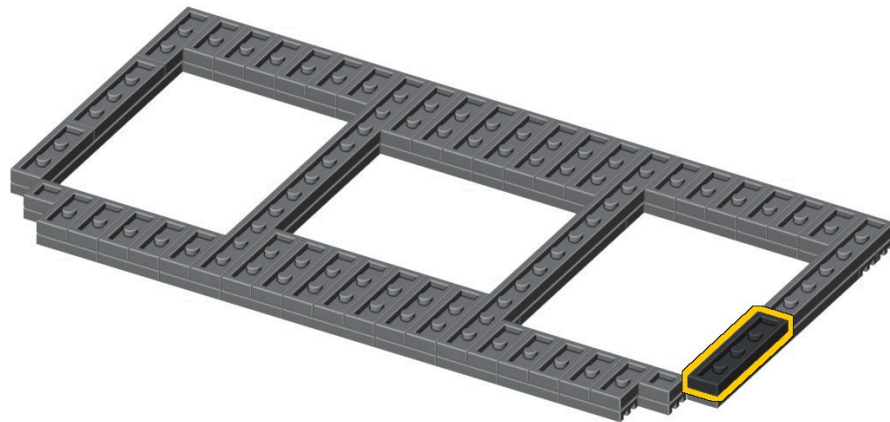
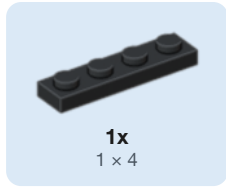
68

Top shell

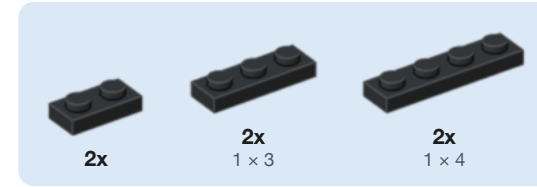


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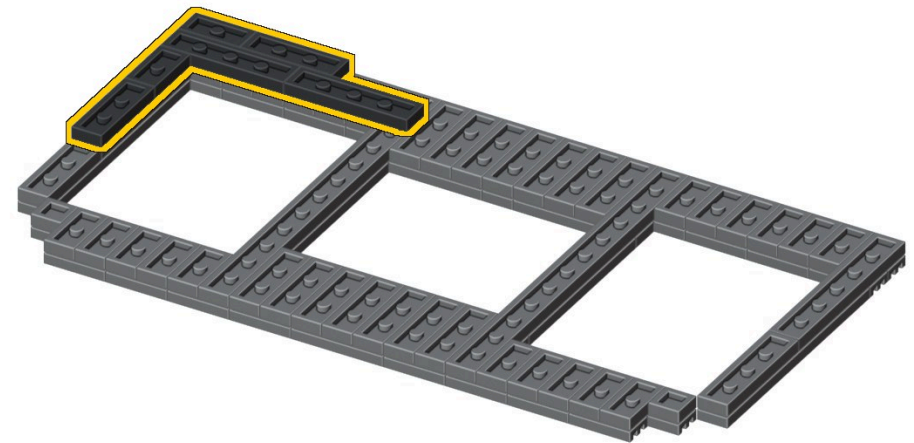
69



70

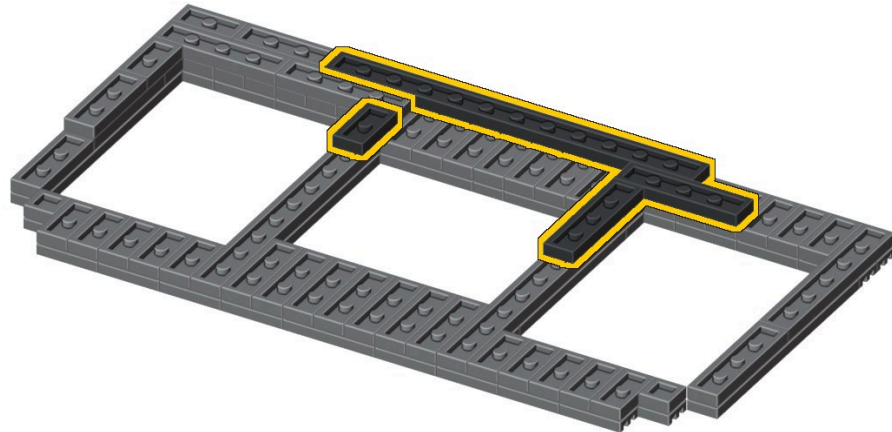
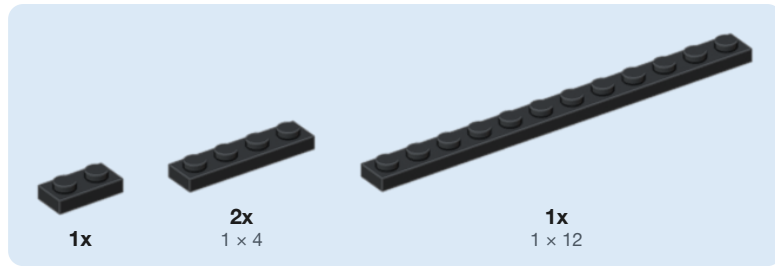


A third layer, with the flap's pivot blocks (plates with a pin hole)



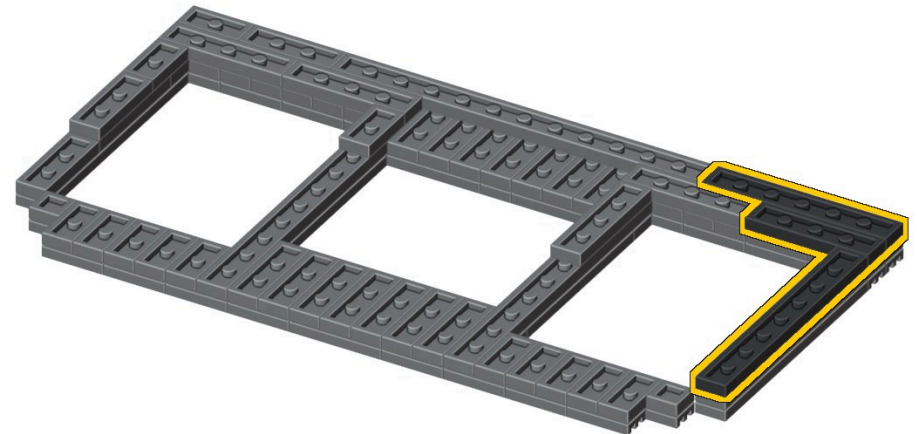
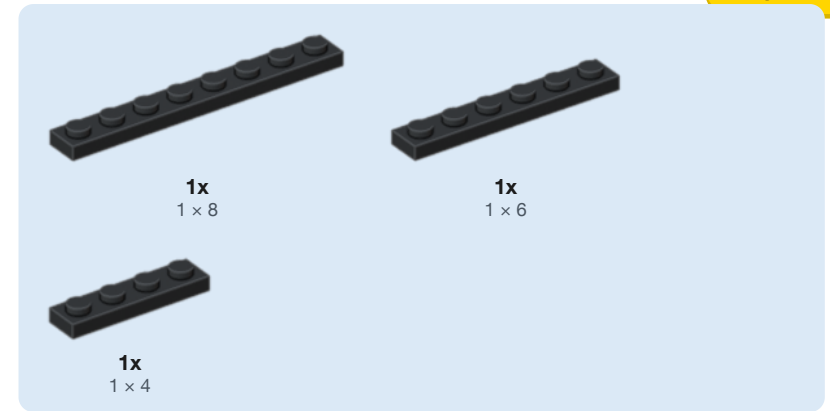
Top shell

71

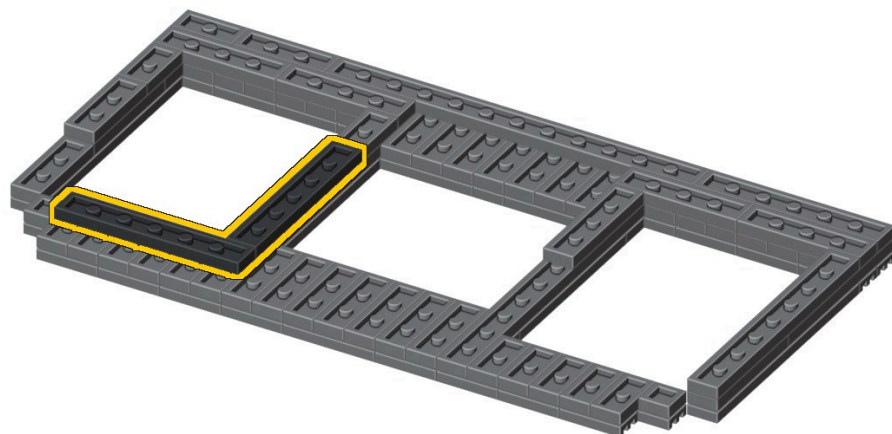
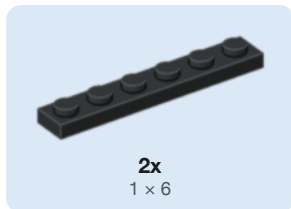


72

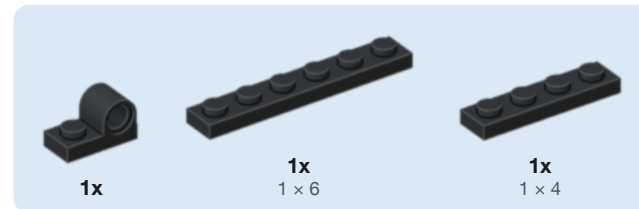
Top shell



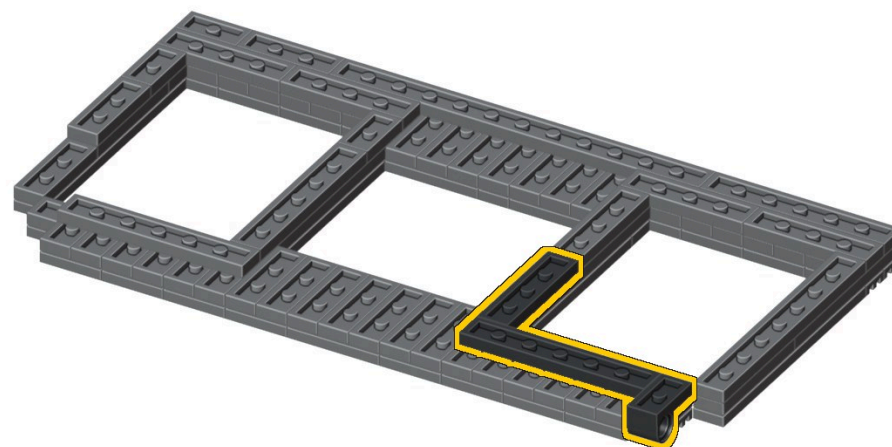
73



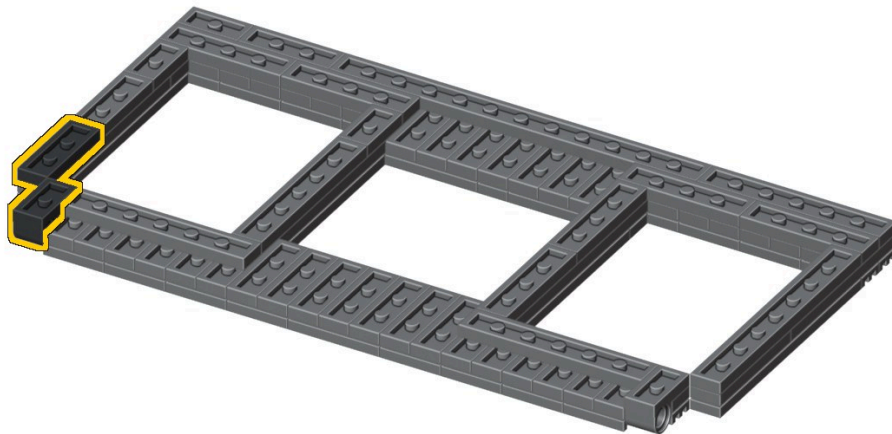
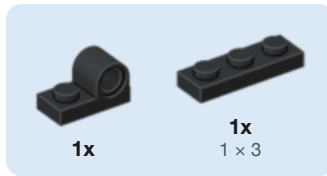
74



Top shell



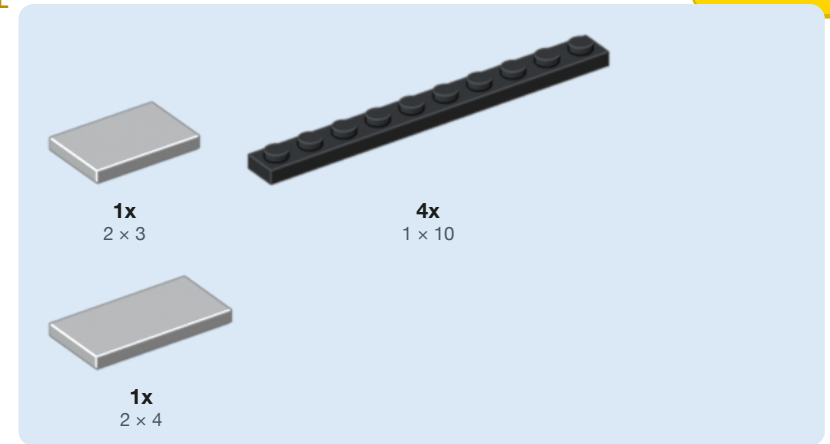
75



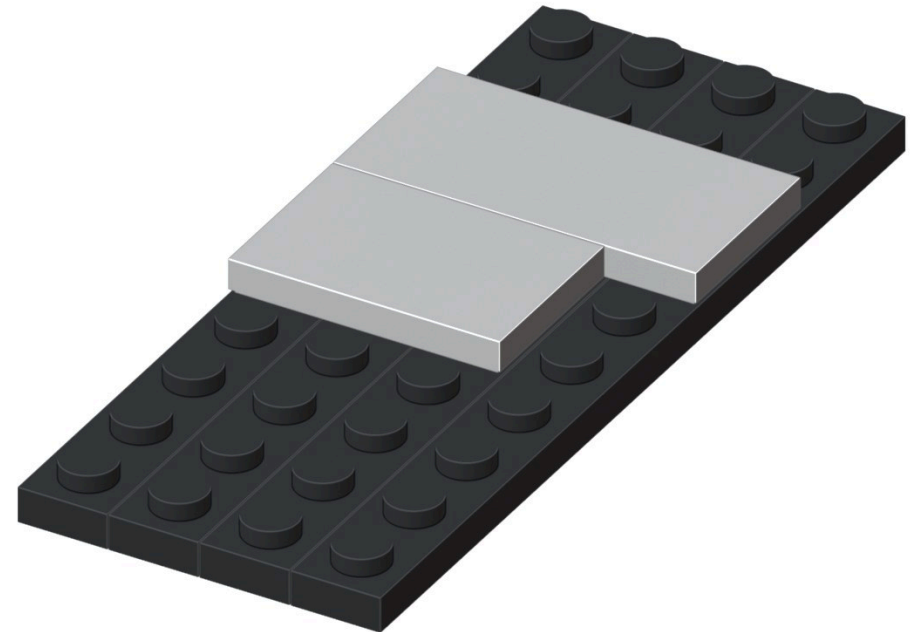
FACE LABEL

76

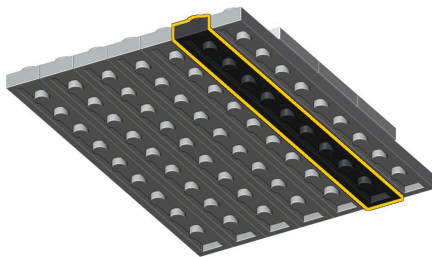
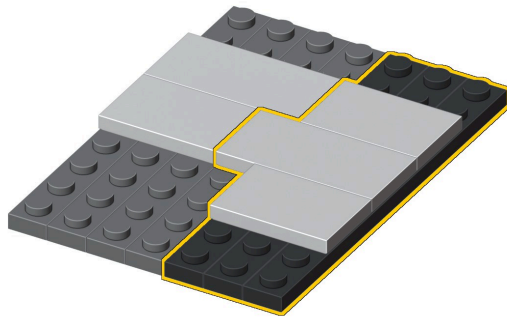
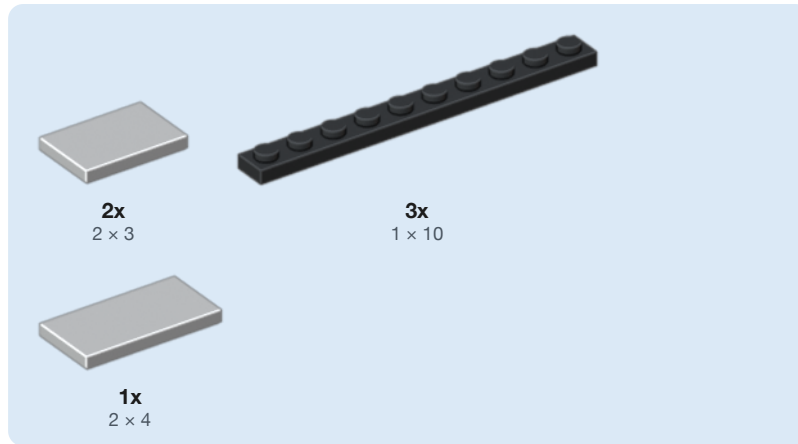
Face label



The face label: black plates running front to back ...

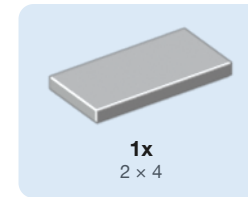


77

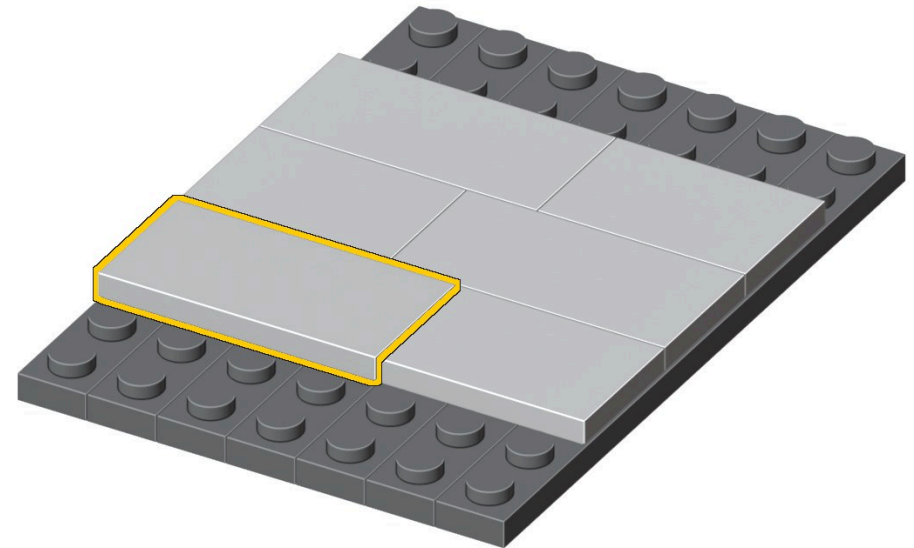


From below: the outlined piece clicks in here.

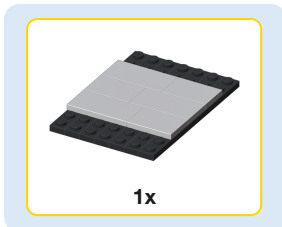
78



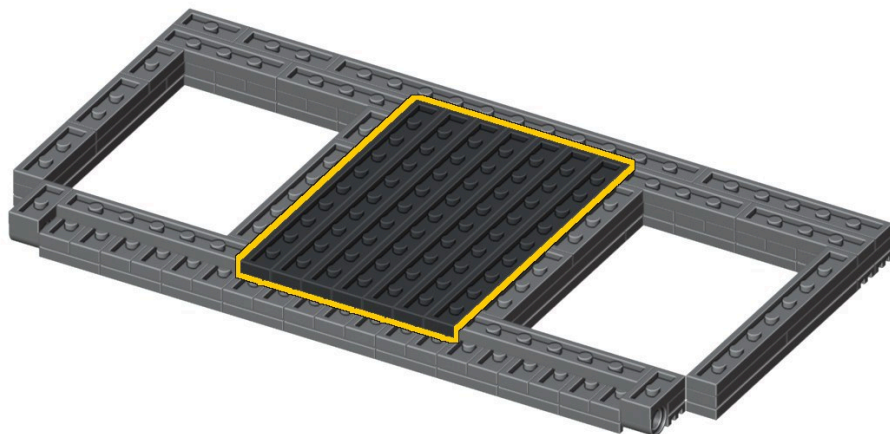
Face label



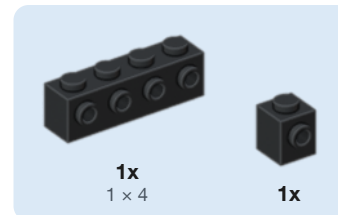
79



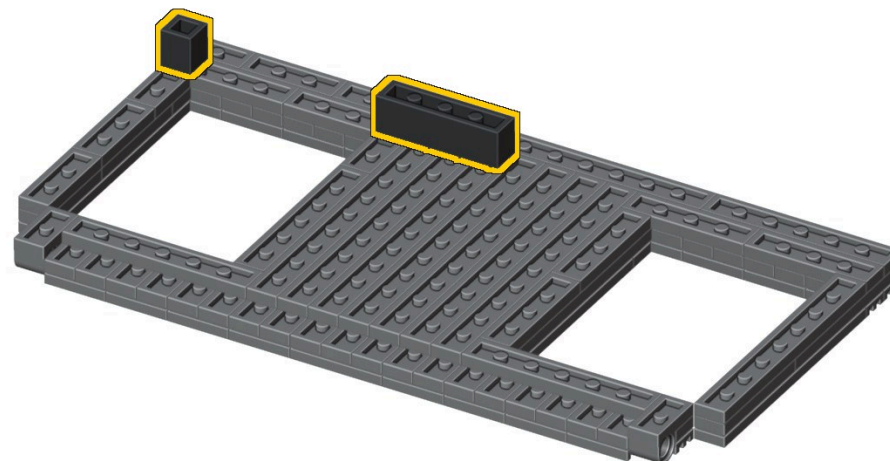
The face label panel: it ties the front and back grip bands together



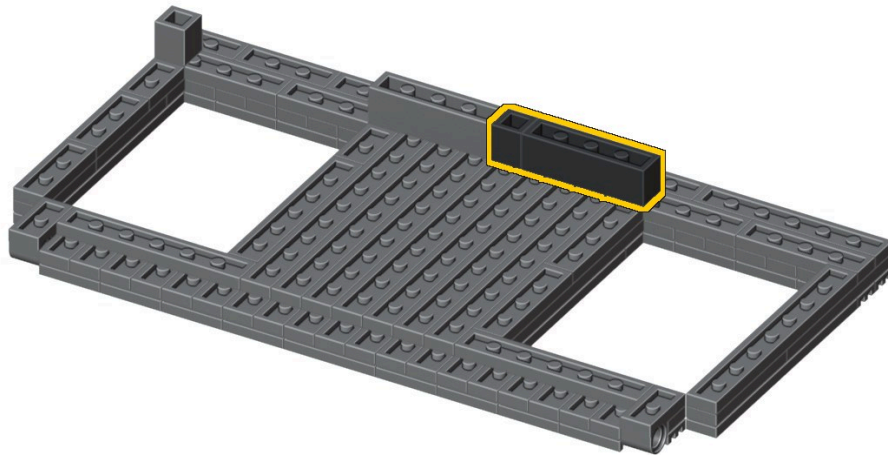
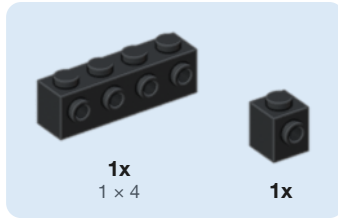
80



Bricks with side studs for the spine label



81



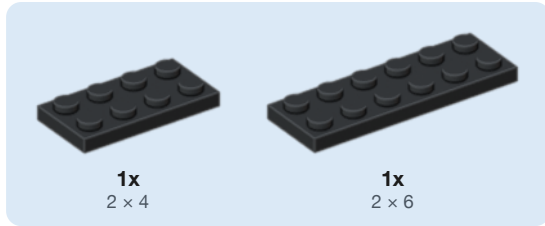
82



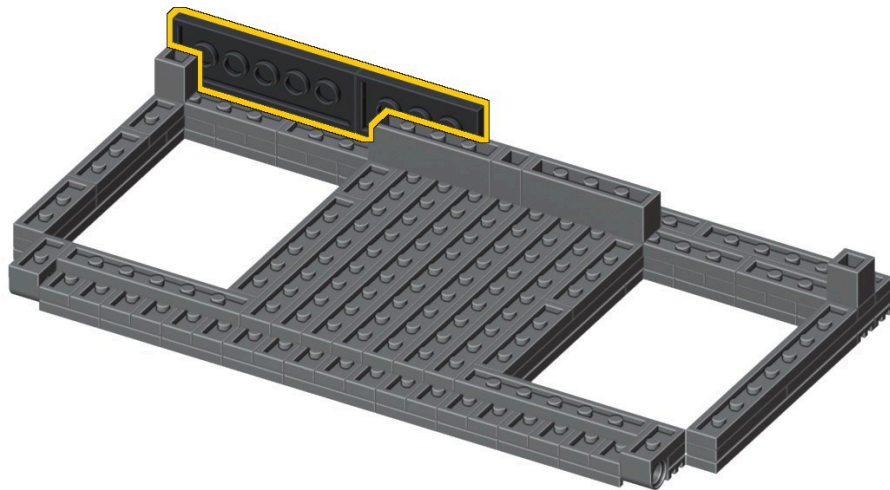
Top shell



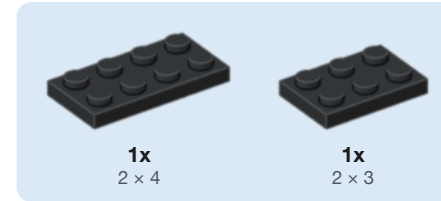
83



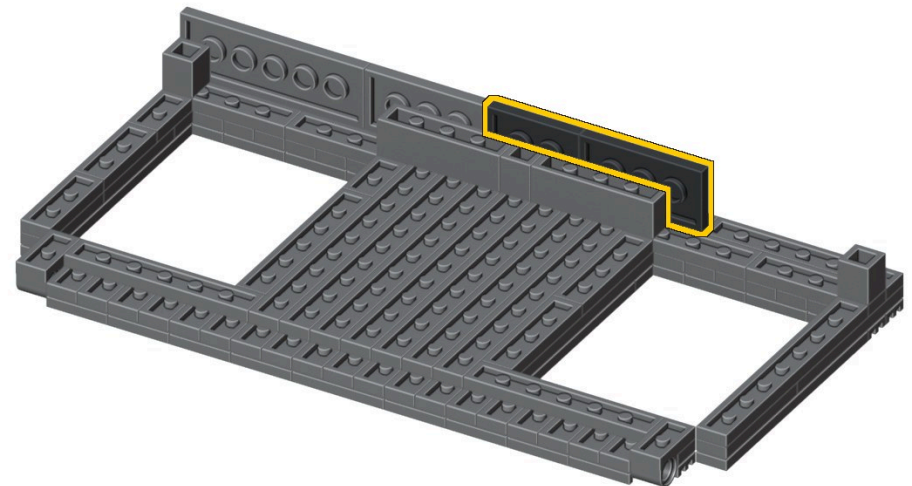
Spine label: plates stood on edge



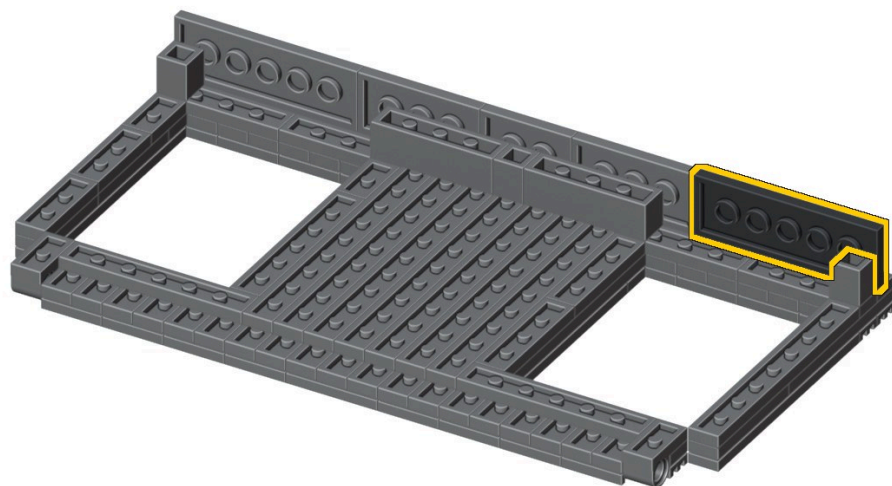
84



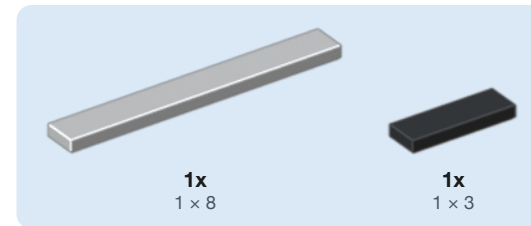
Top shell



85



86



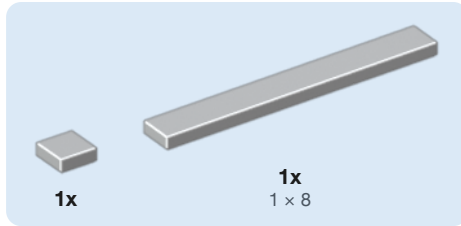
White tiles: the spine label; black tiles at the ends

Top shell

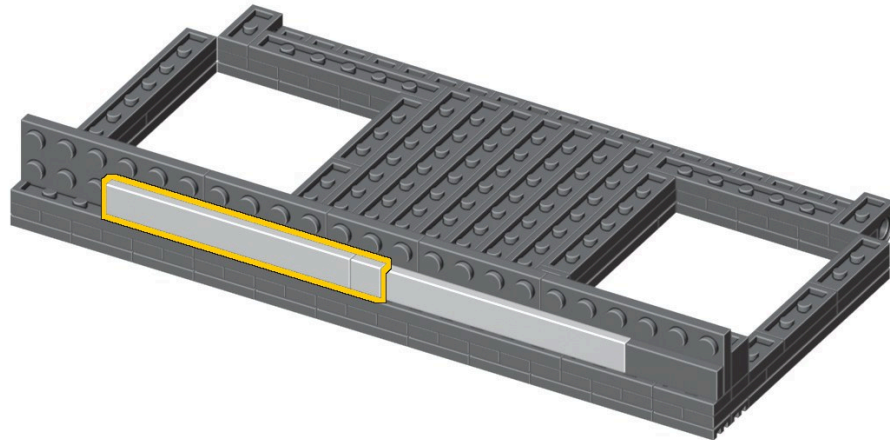
seen from behind



87

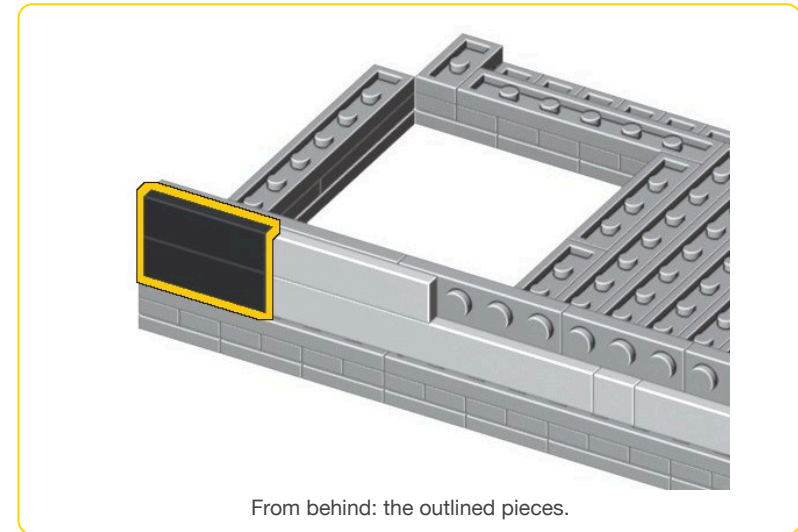
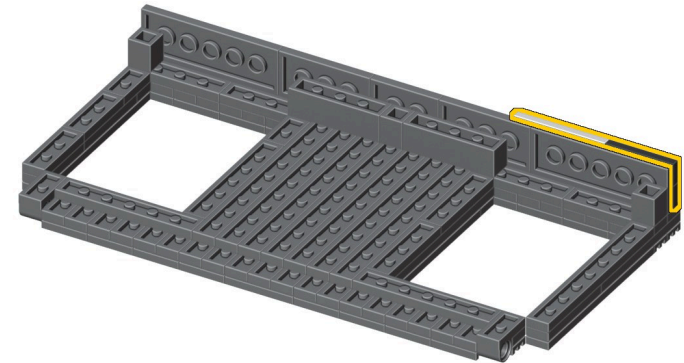
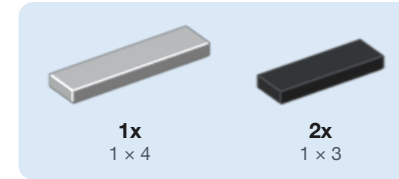


seen from behind

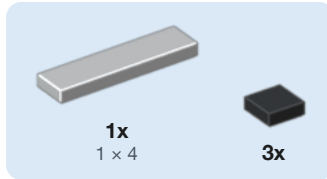


88

Top shell

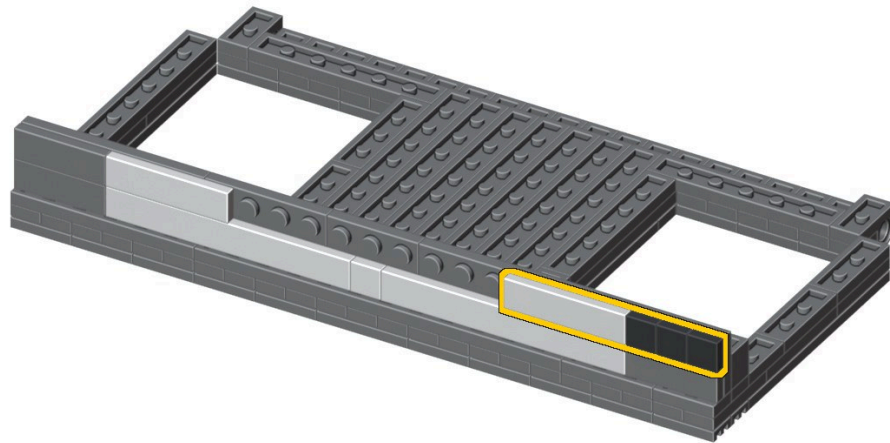


89

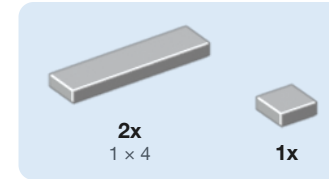


seen from behind

The write-protect tab: a tile you can pull off to protect your recording

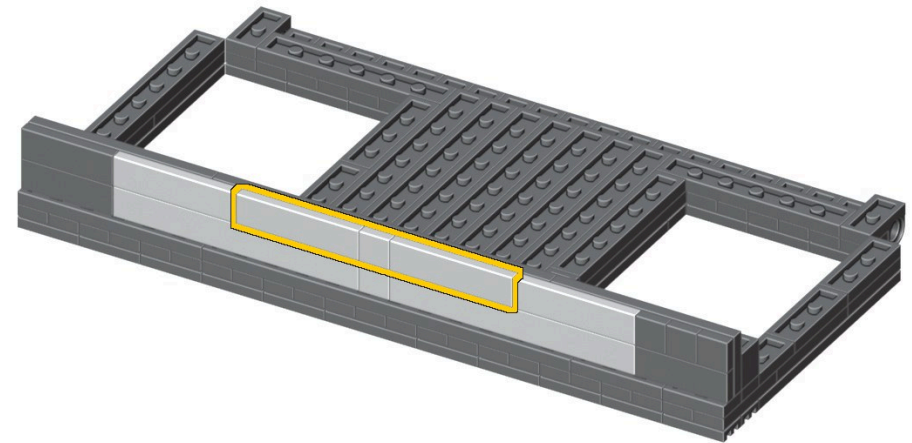


90



seen from behind

Top shell

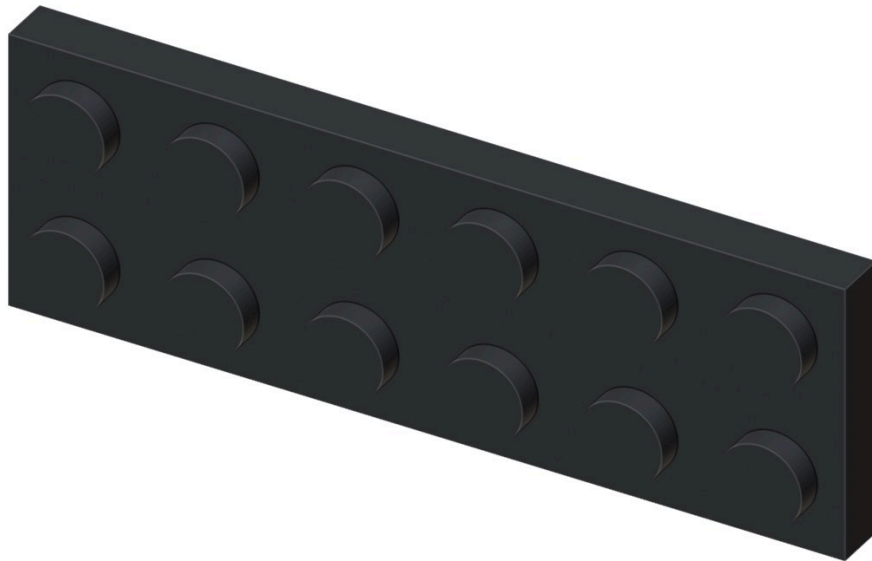


WINDOW · BUILD 2X

91



A window: a 2 x 6 plate stood on its edge, studs to the front



92

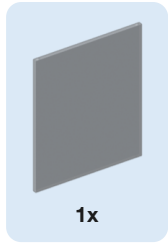


Door frame 1 x 6 x 6, laid flat and pushed onto the plate's top row



Window

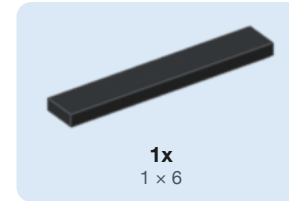
93



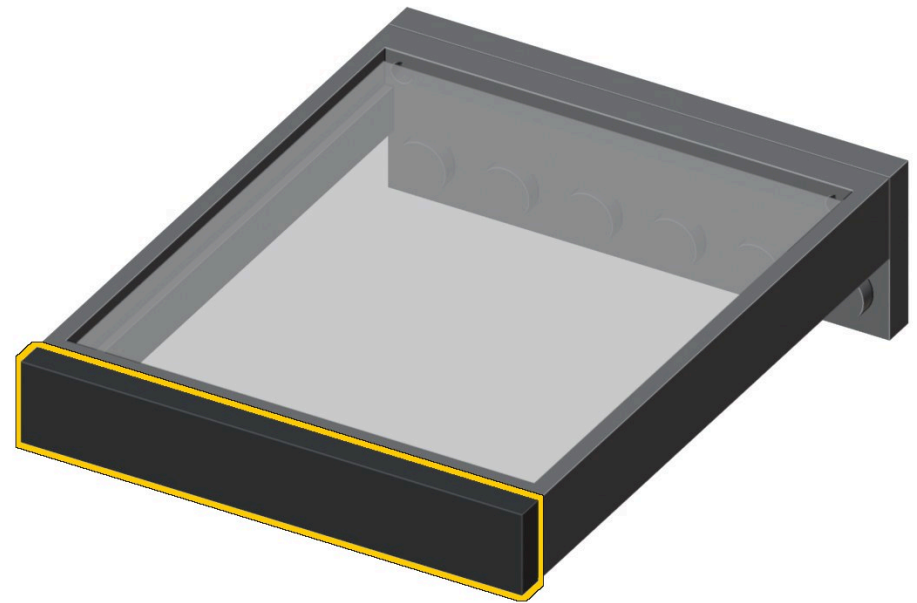
The glass



94



A tile caps the frame's front studs

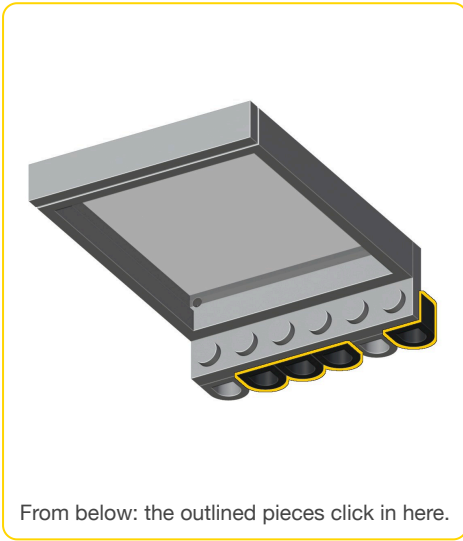
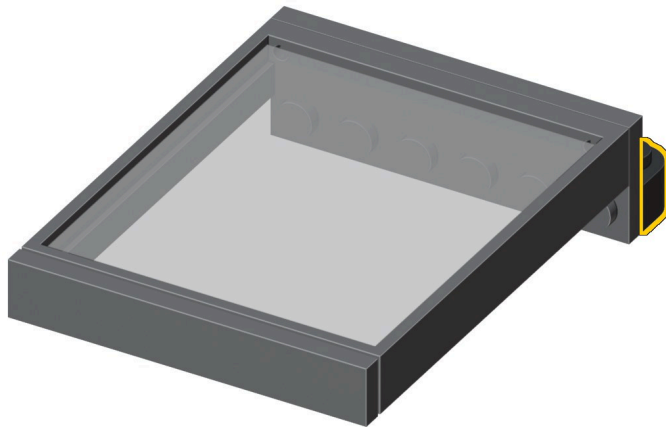


Window

95

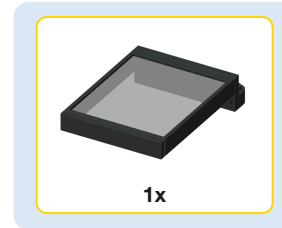


Six hangers on the plate's back row

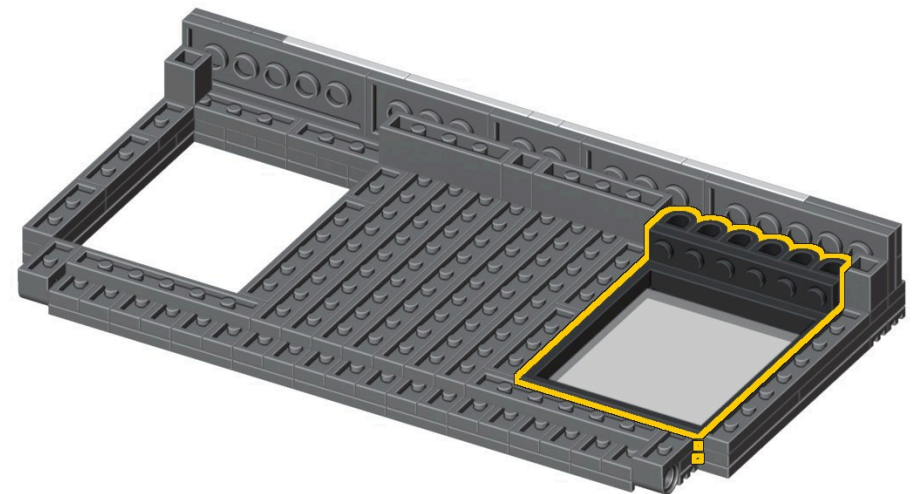


96

TOP SHELL

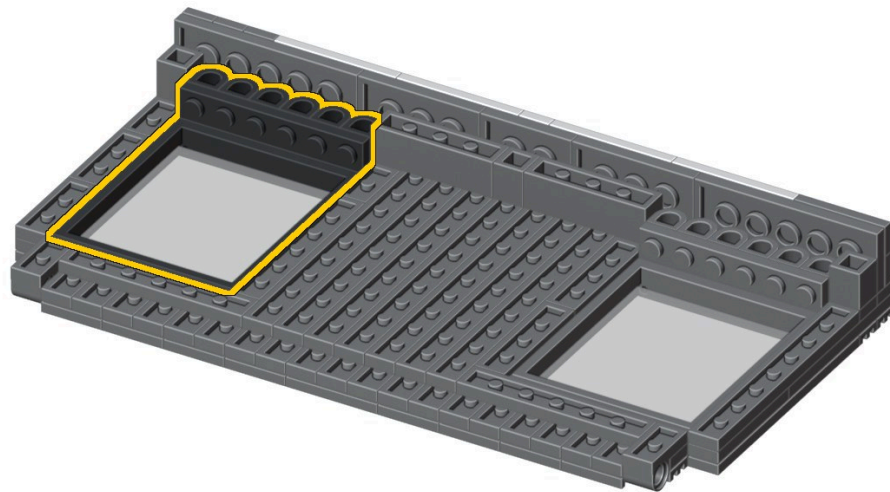
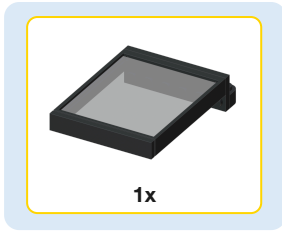


The window, hung on its six hangers



Top shell

97



VHS CASSETTE

98



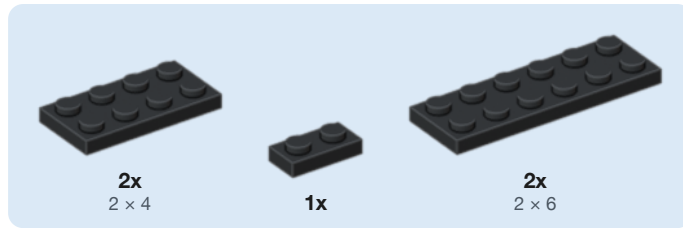
Lower the top shell onto the chassis



VHS Cassette

DUST FLAP

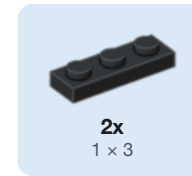
99



Dust flap, built flat: plates for the two lower rows



100

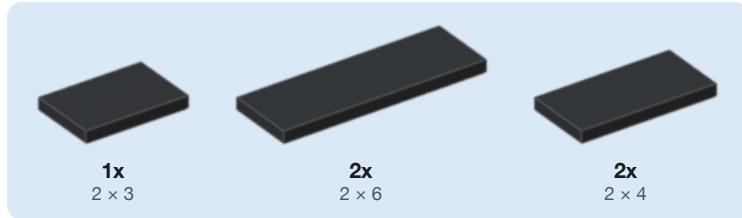


A 1 x 3 plate at each end reaches the top row

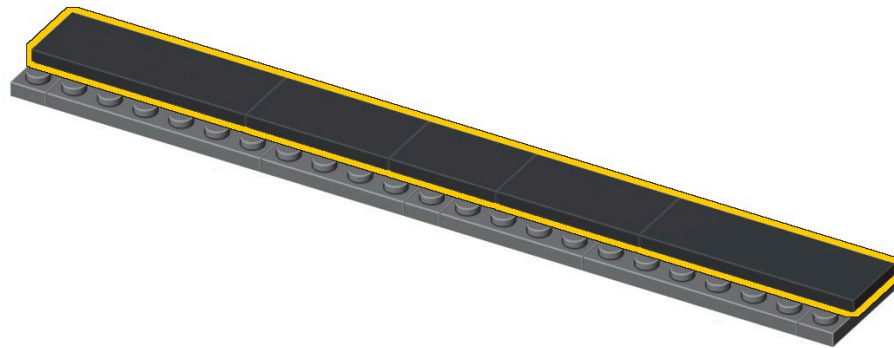


Dust flap

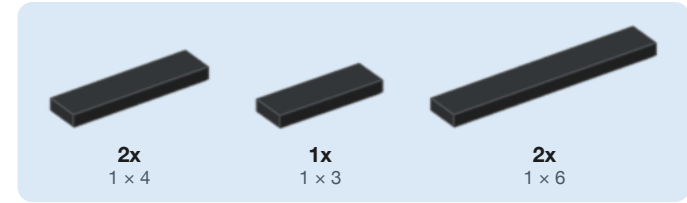
101



2 x N tiles: their top halves hang over the plates and make the flap's top row



102



Bottom row of tiles

Dust flap



103



Turn it over: Technic bricks at the ends are the flap's cheeks

seen from below



VHS CASSETTE

104

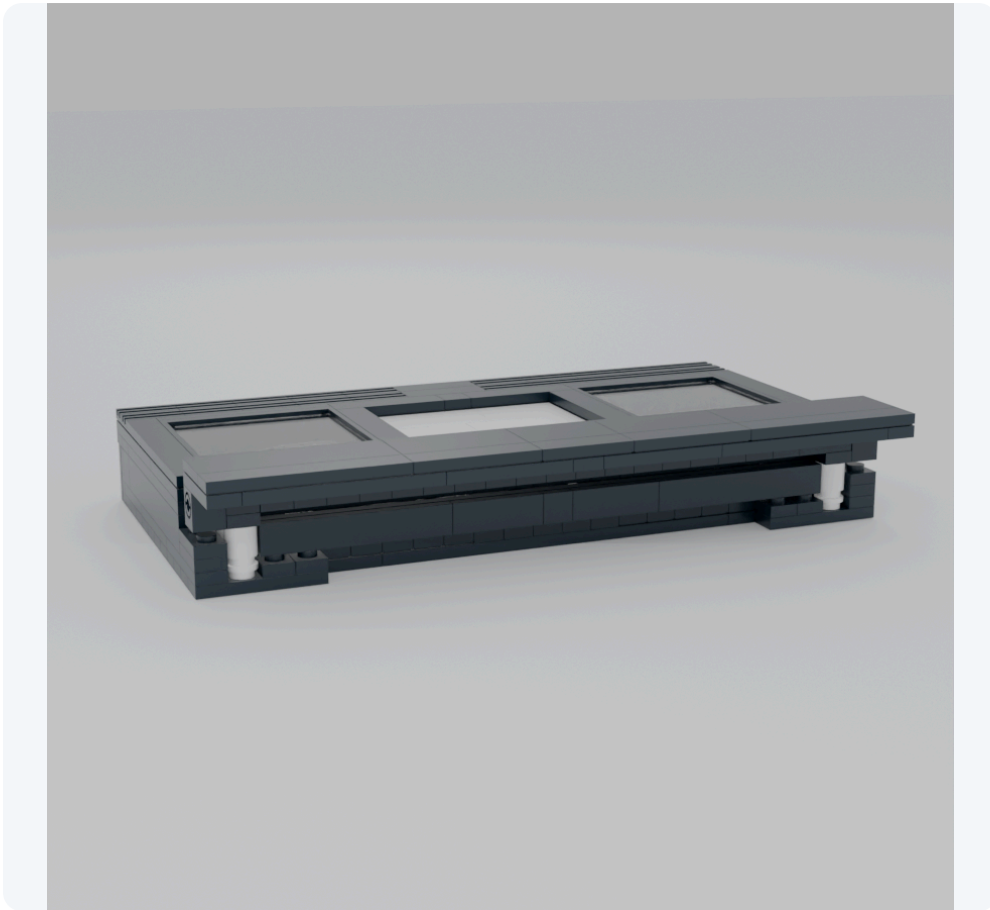
VHS Cassette



Hold the dust flap in place and push a pin through each cheek

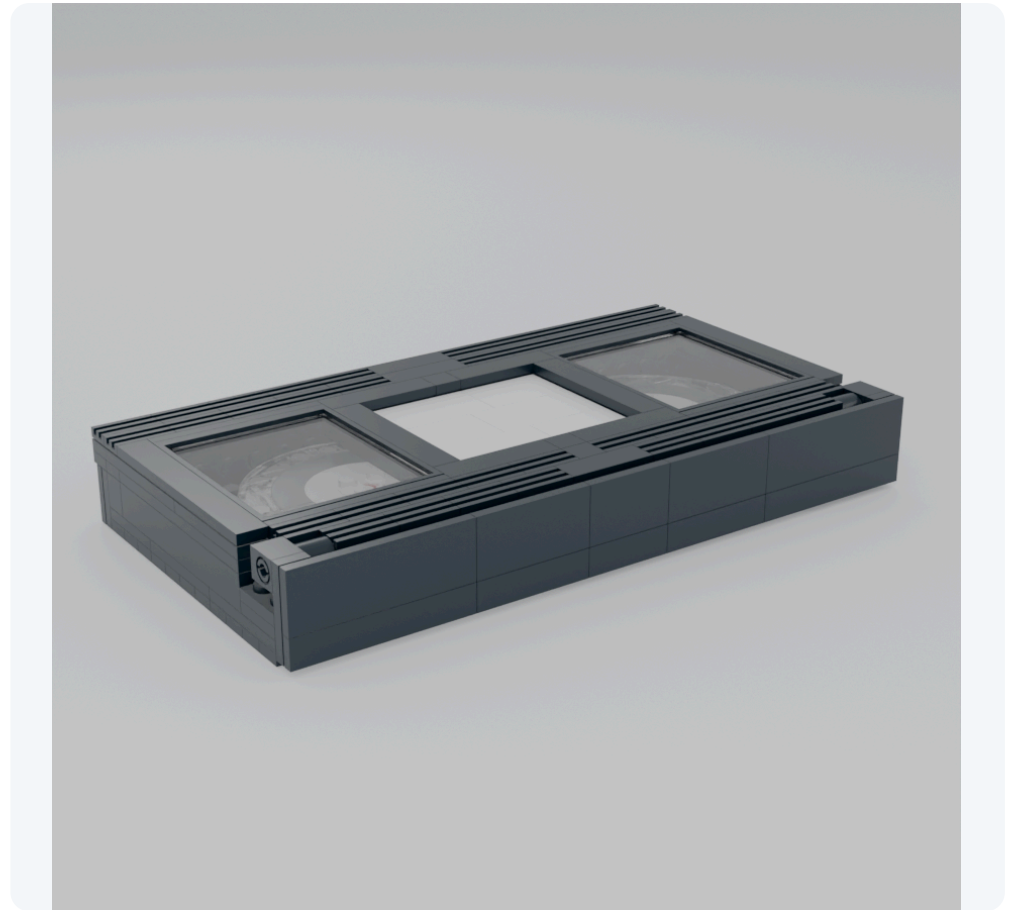


It works!



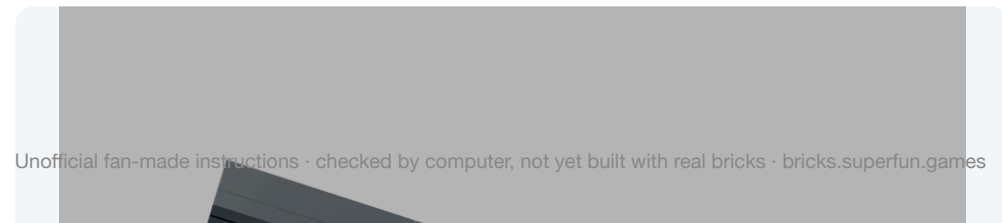
Dust flap

The whole front edge is the flap. It swings up and out 90 degrees on two Technic pins through the end cheeks, uncovering the tape between two white guide rollers.



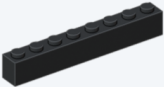
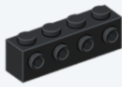
Turning reels

Each reel turns on a red Technic axle in round holes in the chassis. Watch the supply and take-up packs through the clear windows, or turn a spindle from underneath like a VCR does.



Parts inventory

384 pieces in 53 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 2x 300826 Black BrickLink 3008	 1x 306226 Black BrickLink 3062b	 1x 614326 Black BrickLink 3941	 6x 4558954 Black BrickLink 87087	 2x 4162443 Black BrickLink 30414	 2x 6391342 Black BrickLink 42205	 12x 302426 Black BrickLink 3024	 9x 447726 Black BrickLink 4477
 6x 4514845 Black BrickLink 60479	 48x 302326 Black BrickLink 3023	 48x 362326 Black BrickLink 3623	 24x 371026 Black BrickLink 3710	 12x 366626 Black BrickLink 3666	 8x 346026 Black BrickLink 3460	 1x 302126 Black BrickLink 3021	 4x 302026 Black BrickLink 3020
 6x 379526 Black BrickLink 3795	 1x 614126 Black BrickLink 6141	 1x 403226 Black BrickLink 4032a	 4x 6393938 Black BrickLink 74611	 12x 6424674 Black BrickLink 3386	 2x 6114987 Black BrickLink 11458	 2x 654126 Black BrickLink 6541	 2x 4121715 Black BrickLink 2780
 1x 6280168 Black BrickLink 32124	 3x 370926 Black BrickLink 3709	 7x 307026 Black BrickLink 3070b	 4x 306926 Black BrickLink 3069	 21x 4558170 Black BrickLink 63864	 14x 243126 Black BrickLink 2431	 14x 663626 Black BrickLink 6636	 2x 416226 Black BrickLink 4162
 1x 6162892 Black BrickLink 26603	 2x 4560182 Black BrickLink 87079	 2x 6318582 Black BrickLink 69729	 8x 6434275 Black BrickLink 79393	 38x 241226 Black BrickLink 2412b	 2x 6143850 Black BrickLink 15535	 5x 4633691 Flat Silver BrickLink 6141	 2x 4142865 Red BrickLink 32062

Parts inventory (continued)

384 pieces in 53 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

 2x 6511024  Trans-Clear BrickLink 42509	 8x 6544432  Trans-Clear BrickLink 27507	 3x 306201  White BrickLink 3062b	 1x 302301  White BrickLink 3023	 1x 362301  White BrickLink 3623	 3x 614101  White BrickLink 6141	 2x 403201  White BrickLink 4032a	 2x 307001  White BrickLink 3070b
 4x 243101  White BrickLink 2431	 2x 416201  White BrickLink 4162	 8x 6172674  White BrickLink 27925	 3x 6156667  White BrickLink 26603	 3x 4560178  White BrickLink 87079			

About this model

Built like a real cassette, from the inside out. First the chassis: the bottom with its drive holes and screw heads, the lamp tube, the reel-lock release, the guide rollers and the tape stretched across the mouth. Two reels drop onto their red spindles: the supply reel and the take-up reel. The end walls go on, then the top shell, built upside down with its grip bands, recessed label, spine label and two big windows, is lowered over everything. Last, the full-length dust flap hangs on two Technic pins.

Checked on a computer

PASS	real elements	53 part/colour combinations: 0 not real, 0 rare
PASS	connections	996 connections (965 stud, 21 pin, 6 axle, 4 hinge); 1 separate piece(s)
PASS	collisions	0 overlapping pair(s) among 384 parts
PASS	buildability	283 insertions tried across 8 submodel(s); 0 problem(s)
PASS	stability	about 313 g; centre of mass inside the base (98% margin); tips at 74.9 deg
PASS	mechanism	24 poses swept for 3 moving group(s); 0 gear pair(s); 0 problem(s)
PASS	electrics	no electrics in this model
PASS	technique	0 note(s)

Not yet built with real bricks

These checks catch most mistakes a designer can make, but not all: clutch power, part tolerances, how stiff a long axle is and how the model survives handling can only be judged with real bricks. If you build it, we would love to hear how it went.

Files

3D model (LDraw, opens in BrickLink Studio, LeoCAD and LDCad), parts lists for BrickLink and LEGO Pick a Brick, and an interactive 3D viewer:

bricks.superfun.games/m/vhs_tape

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